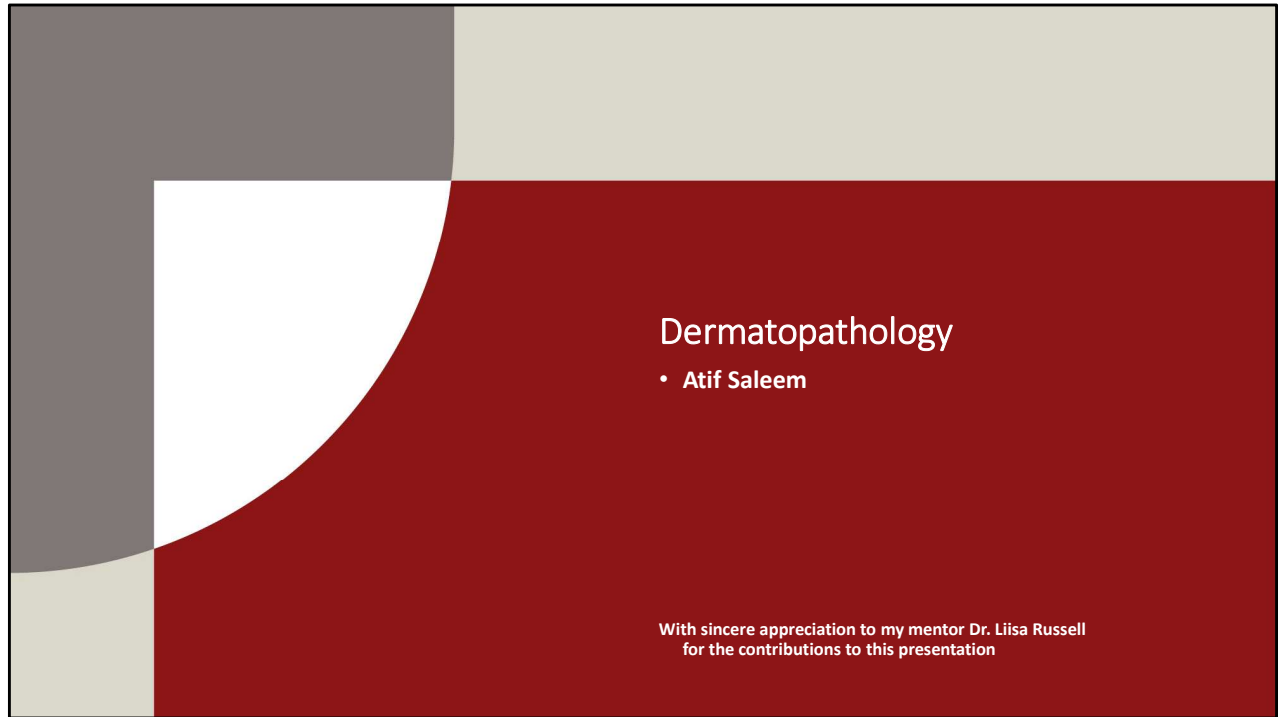




Objectives

Focus on red highlighted items throughout the presentation

Benign epithelial tumors and tumor-like lesions:

Know the relative incidence and recognize and describe the pathophysiologic and diagnostic features of common benign epithelial tumors and tumor-like lesions:

1. Virally induced epithelial proliferations
 - a. Molluscum contagiosum
 - b. Verruca vulgaris and related lesions
2. Fibroepithelial polyp

Acrochordon = squamous papilloma = skin tag
3. **Seborrheic keratosis** (Sign of Leser-Trelat = sudden eruption of numerous SKs; associated with internal carcinomas)
4. Dermatositis papulosa nigra
5. Epithelial cyst (wen)
6. Acanthosis nigricans

Premalignant and malignant epidermal lesions:

Describe the relative frequency, risk factors, pathogenesis, pathophysiologic and morphologic features and factors affecting prognosis for

1. **Actinic keratosis**
2. **Squamous cell carcinoma**
3. **Keratoacanthoma**
4. **Basal cell carcinoma**

Objectives

Adnexal tumors and tumor-like lesions:

Recognize //(list) the most common tumors and tumor-like lesions of skin appendages

1. Sebaceous hyperplasia
2. Sebaceous adenoma

Muir-Torre syndrome: multiple sebaceous adenomas/sebaceous carcinomas associated with hereditary nonpolyposis colorectal carcinomas

3. Sebaceous carcinoma
4. Cylindroma ("turban tumor")
5. Trichoepithelioma
6. **Pilomatrixoma**
7. Trichilemmoma; association with Cowden syndrome
8. Apocrine carcinoma

Tumors of the dermis:

Describe the relative frequency, risk factors, pathogenesis and pathophysiologic and morphologic features of:

1. **Benign dermatofibroma (benign fibrous histiocyoma)**
2. Dermatofibrosarcoma protuberans

Objectives

Disorders of pigmentation and melanocytes:

1. Ephelids (freckle) vs. lentigo vs. **solar lentigo** vs. **melasma**
2. Melanocytic nevi

Describe the stages of development, and pathophysiologic and diagnostic features of melanocytic nevi:

- a. **Junctional nevus**
- b. **Compound nevus**
- c. **Intradermal nevus**

3. Describe variants of melanocytic nevi and their clinical significance:

- a. Congenital nevus
- b. Blue nevus
- c. Spitz nevus (spindle and epithelioid cell nevus)
- d. Halo nevus
- e. Dysplastic (Clark) nevus
 - Describe dysplastic nevus syndrome

4. **Malignant melanoma**

- a. **Describe the risk factors and pathogenesis**

Environmental factors
Genetic variation

Mutations

- b. **Recognize and describe the pathophysiologic and morphologic features of malignant melanoma**

Radial growth phase:

Lentigo maligna melanoma
Superficial spreading melanoma
Acral/mucosal lentiginous melanoma

Vertical growth phase:

Nodular melanoma

- c. **Describe prognostic evaluation in melanoma:**

Clark's levels

Breslow's measurements of melanoma thickness

5. **Compare and contrast the key diagnostic gross and histologic features of benign melanocytic compound nevus vs. dysplastic nevus vs. malignant melanoma**

Objectives

Cellular migrants to the skin:

1. **Describe the clinical presentation, morphologic features, biologic behavior and prognosis of cutaneous T-cell lymphoma (mycosis fungoides)**

Describe:

- a. **Sezary-Lutzner cells**
- b. **Pautrier microabscesses**

2. Describe the pathogenesis, typical clinical presentation and morphologic features in children and adults of mastocytosis

Describe:

- a. Darier sign
- b. Dermatographism

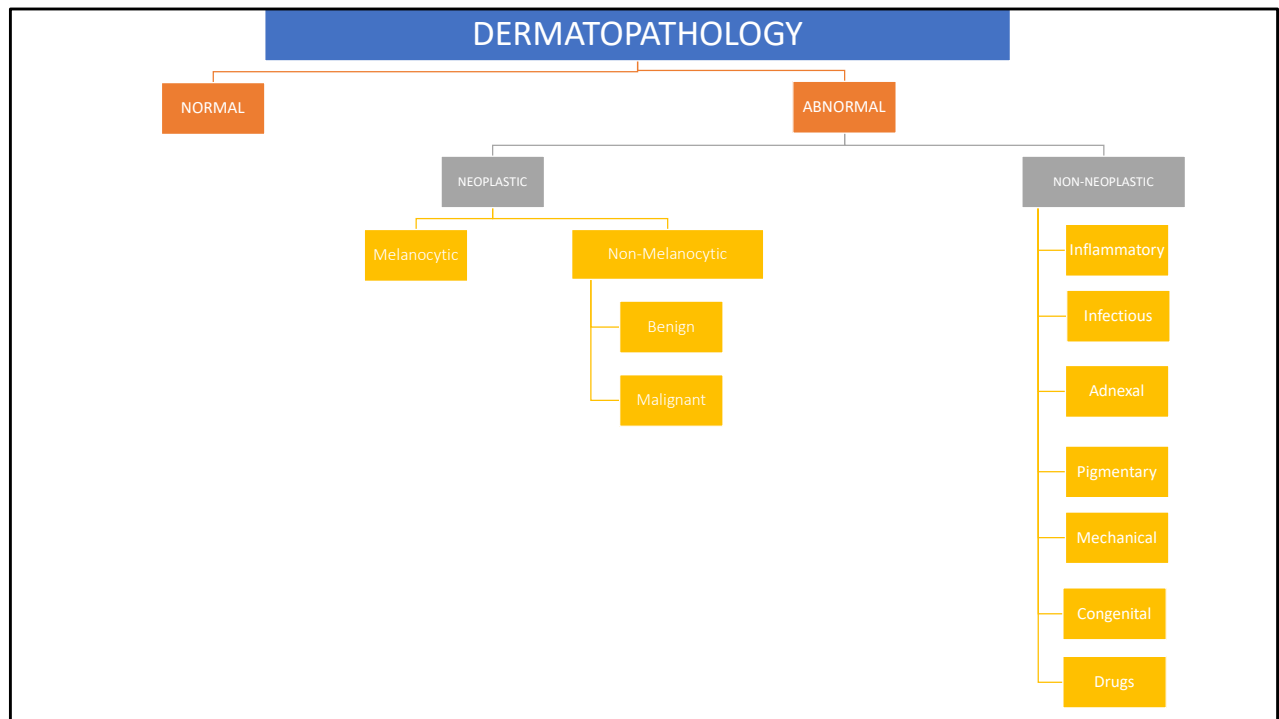
Familial cancer syndromes with cutaneous manifestations:

List the most common inherited cancer syndromes, the inheritance pattern and the types of skin lesions in each:

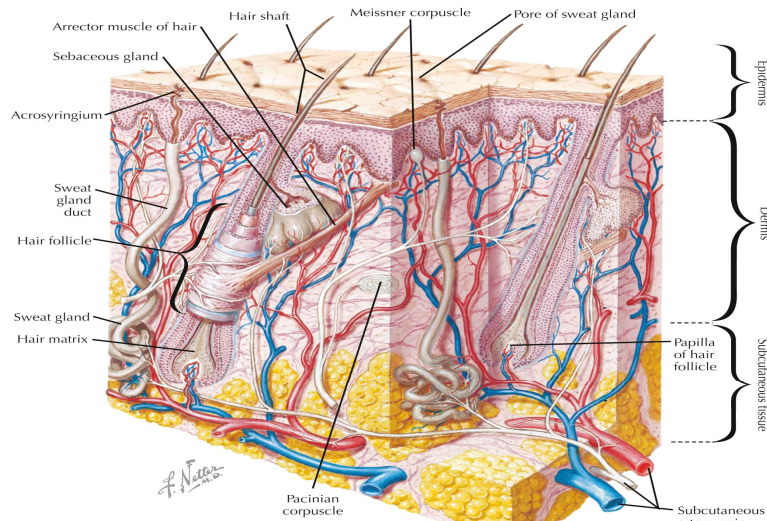
Ataxia-telangiectasia, Cowden syndrome, Familial melanoma syndrome, Muir-Torre syndrome, NF1, NF2, Nevoid basal cell carcinoma syndrome, Peutz-Jeghers, Tuberous sclerosis, Xeroderma pigmentosum

Odds and ends:

1. Melasma versus vitiligo (Compare and contrast the etiology and clinical presentation)



NORMAL



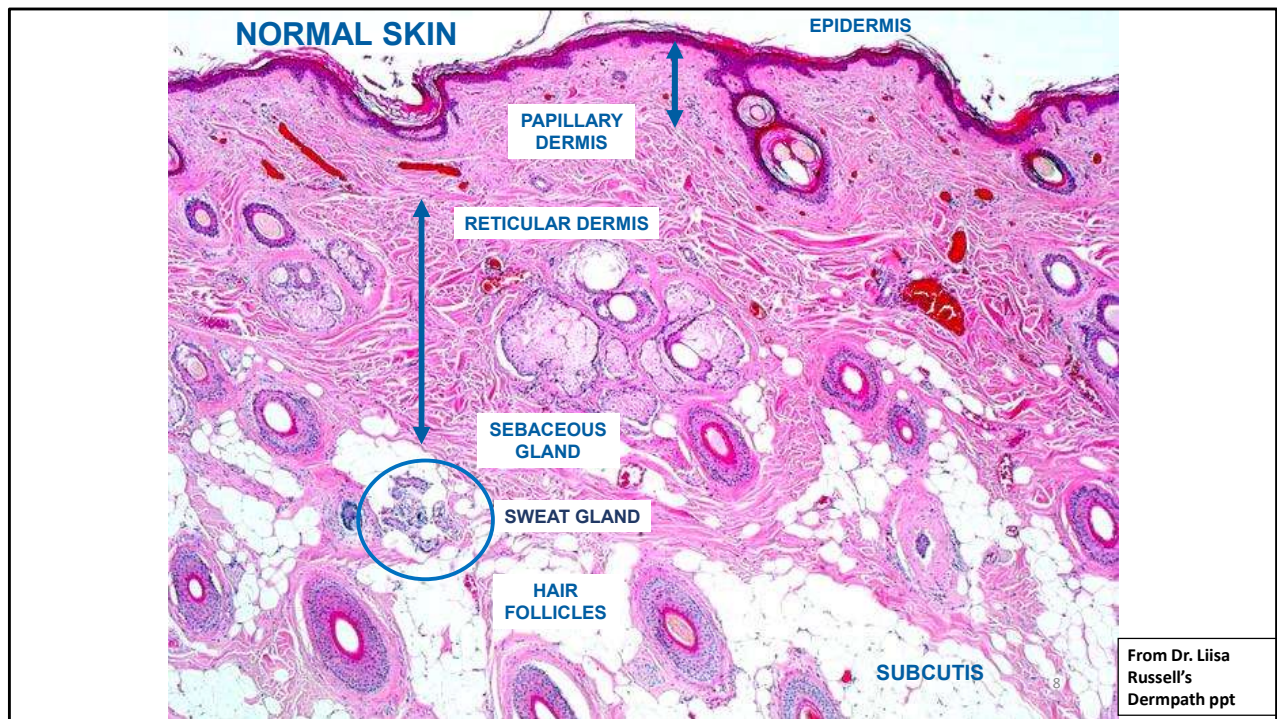
▲ Schematic of skin and its appendages that shows the epidermis, dermis, and subcutaneous tissue.

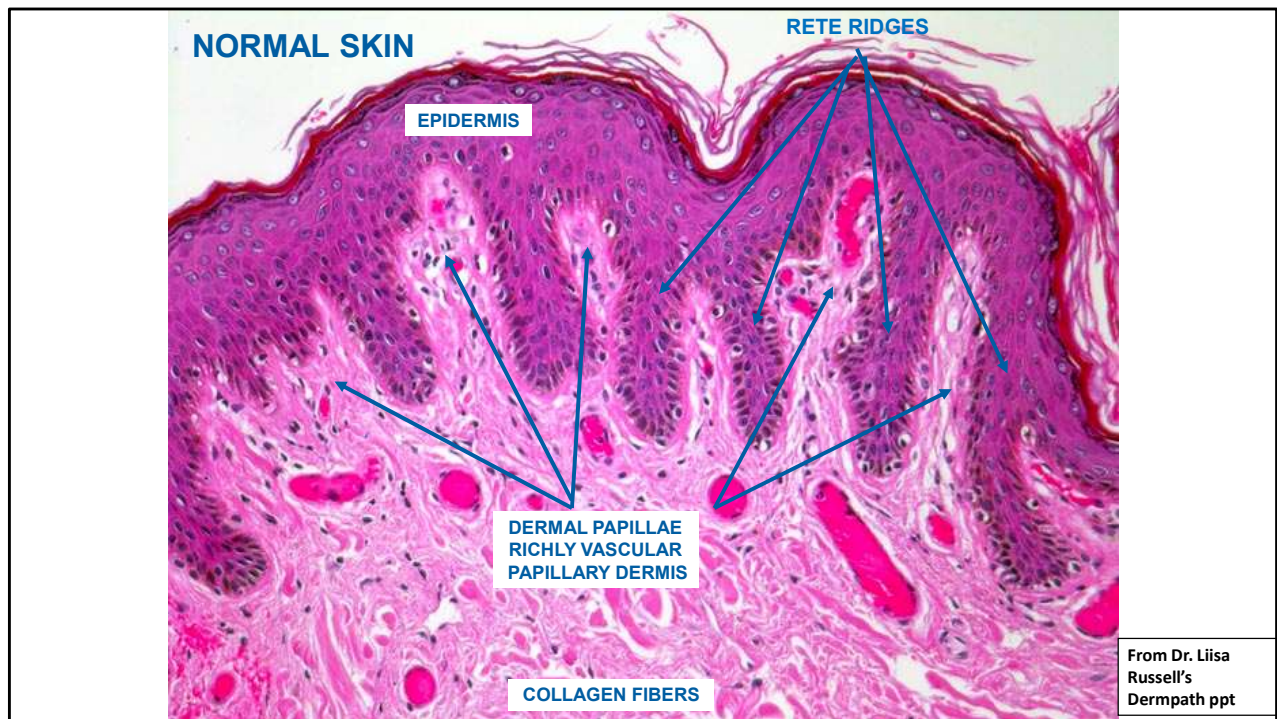
Integumentary System

Ovalle, William K., PhD, Netter's Essential Histology, 11, 263-283

(no summary available)

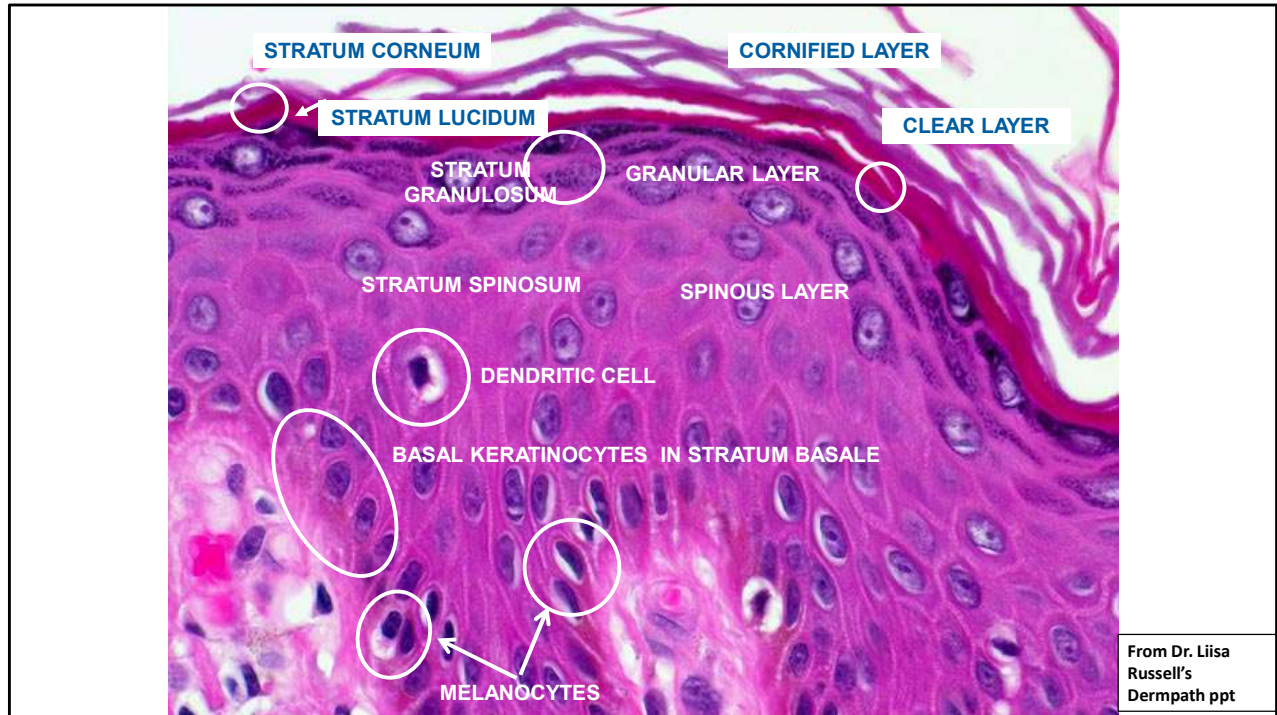
Copyright © 2021 Copyright © 2021 by Elsevier, Inc. All rights reserved.



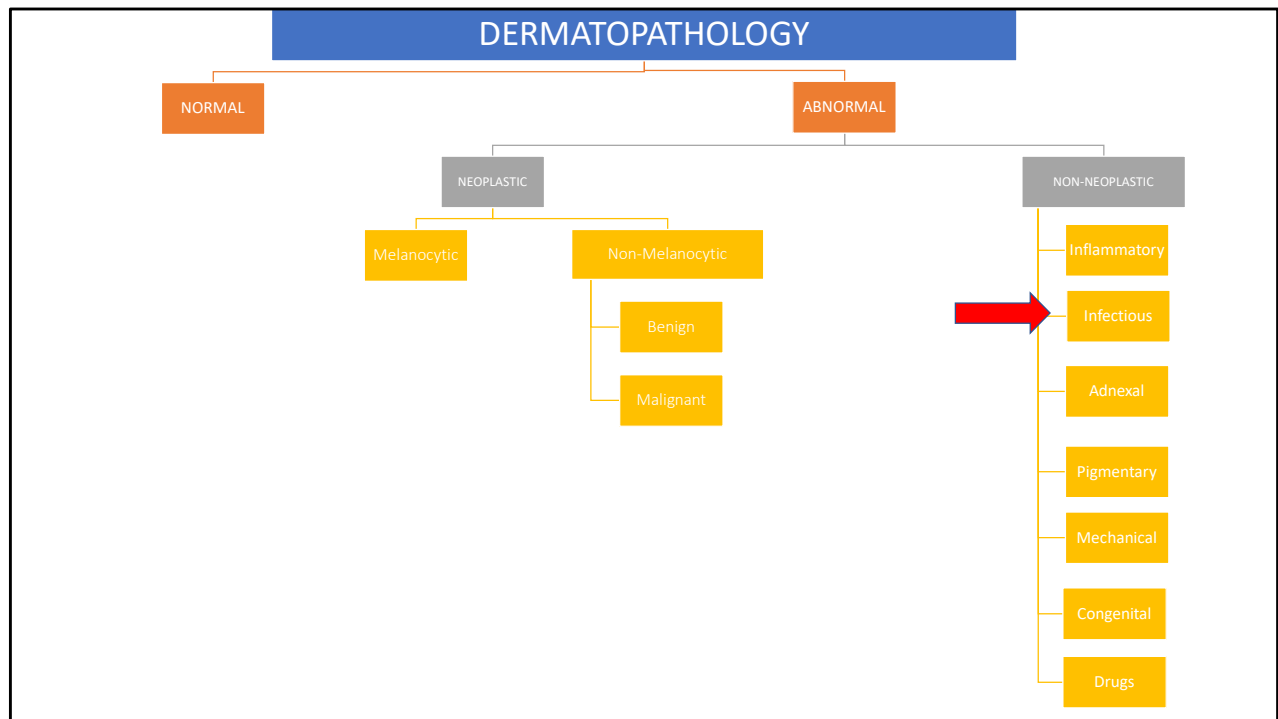


Epidermis and papillary dermis.

normal histology



thick 5, thin 3 layers

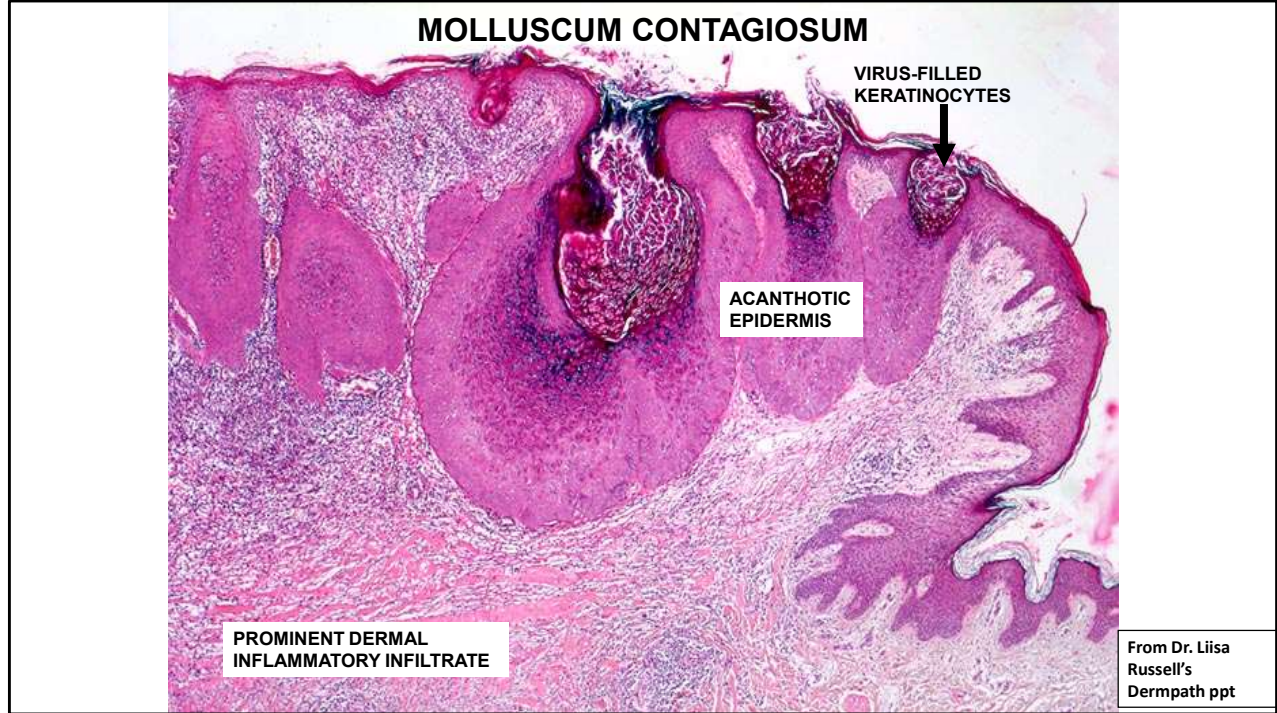


VIRALLY INDUCED PROLIFERATIVE LESIONS: MOLLUSCUM CONTAGIOSUM

From Dr. Liisa
Russell's
Dermopath ppt

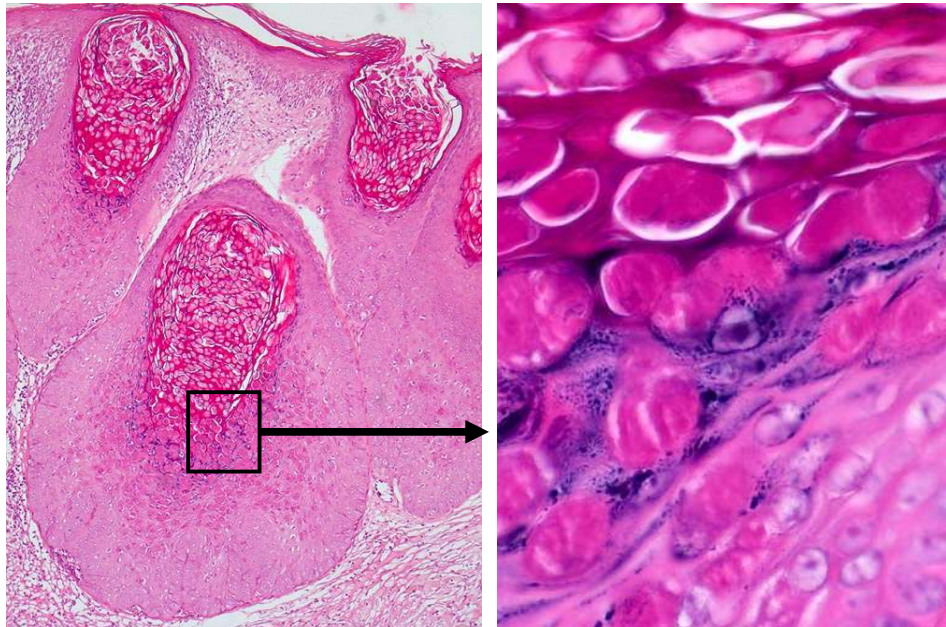


- **DNA Pox virus:** 4 types of MCV, MCV-1 to -4.
- **Infects squamous cells.**
- Cannot be cultured.
- MCV-1 most common in human infections.
- MCV-2 usually in adults and often sexually transmitted.
- Transmission by direct skin contact.
- Autoinoculation common.
- Vertical transmission is likely.
- Firm, smooth, umbilicated, flesh-colored, white, translucent, or yellow papules, 1-15 mm in diameter in groups on skin and mucosal surfaces.
- May be widely disseminated with hundreds of lesions.
- Generally self-limited with resolution in 6-12 months but can persist for several years.
- Destruction of the lesion stops the infection.
- No therapy necessary in immunocompetent people.
- In immunocompromised patients aggressive therapy is needed to contain the disease (cryo-, laser- or chemical therapy, or curettage).



Diagnosis: Molluscum contagiosum. **Description:** Virus-infected epidermis is thickened and acanthotic and has crater-like invaginations filled with virus-infected cells. The infected squamous cells have ball-like intracytoplasmic viral inclusions. **Clinical findings:** Perianal small, linear skin-colored papules. **Stain:** H&E.

MOLLUSCUM CONTAGIOSUM
PINK-STAINING VIRUS PARTICLES INSIDE KERATINOCYTES



From Dr. Liisa
Russell's
Dermopath ppt

Diagnosis: Molluscum contagiosum. **Description:** Virus-infected epidermis is thickened and acanthotic and has crater-like invaginations filled with virus-infected cells. The infected squamous cells have ball-like intracytoplasmic viral inclusions. **Clinical findings:** Perianal small, linear skin-colored papules. **Stain:** H&E.

VIRALLY INDUCED PROLIFERATIVE LESIONS:

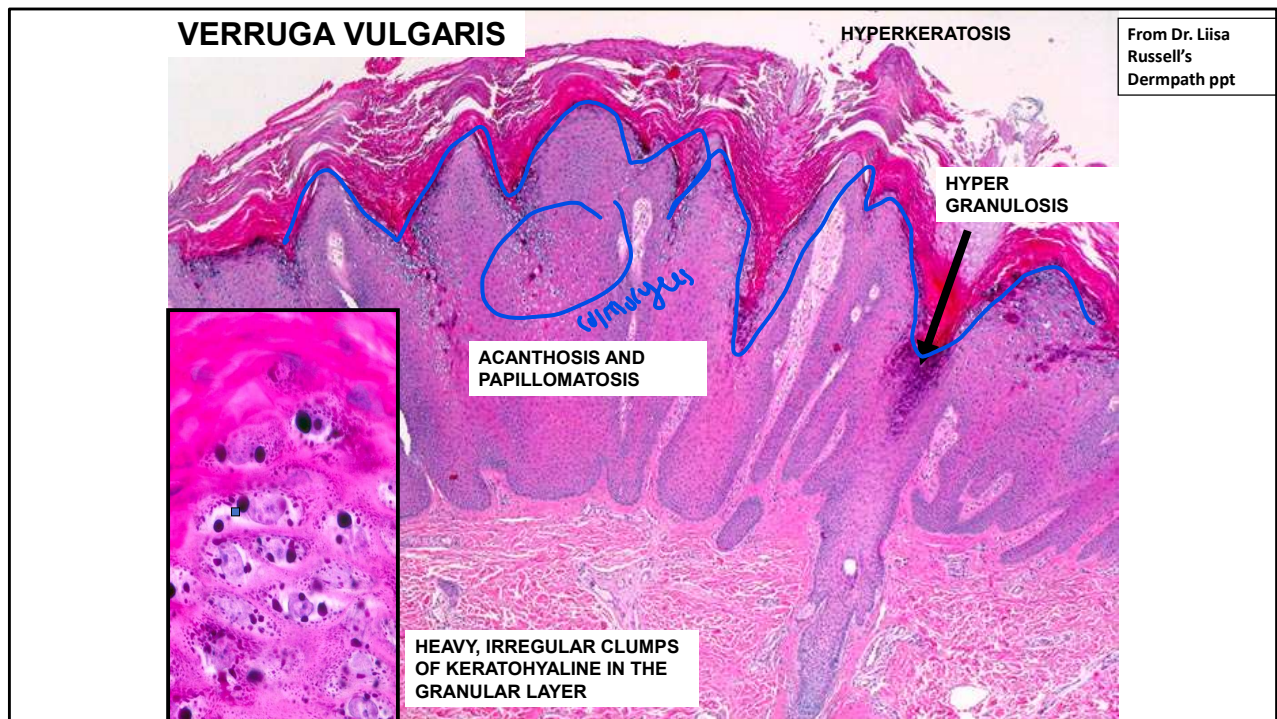
From Dr. Liisa Russell's
Dermopath ppt

VERRUGA VULGARIS



- Warts are caused by the human papilloma virus (HPV), which enters the skin through a cut or scratch.
- HPV infects the squamous epithelium of the skin or genitals.
- 130 known types of HPV exist; each type is typically able to infect only a specific area on the body.
- Infection causes a benign papilloma (wart).
- Warts are the most common dermatological complaint after acne.
- 75% of people will develop a wart (verruca vulgaris) at some time in their lives.
- Warts are slightly contagious, but not everyone exposed to HPV develops a wart.
- Infection can occur at any age, but children and young adults are more prone to it because their defense mechanisms may not be fully developed.

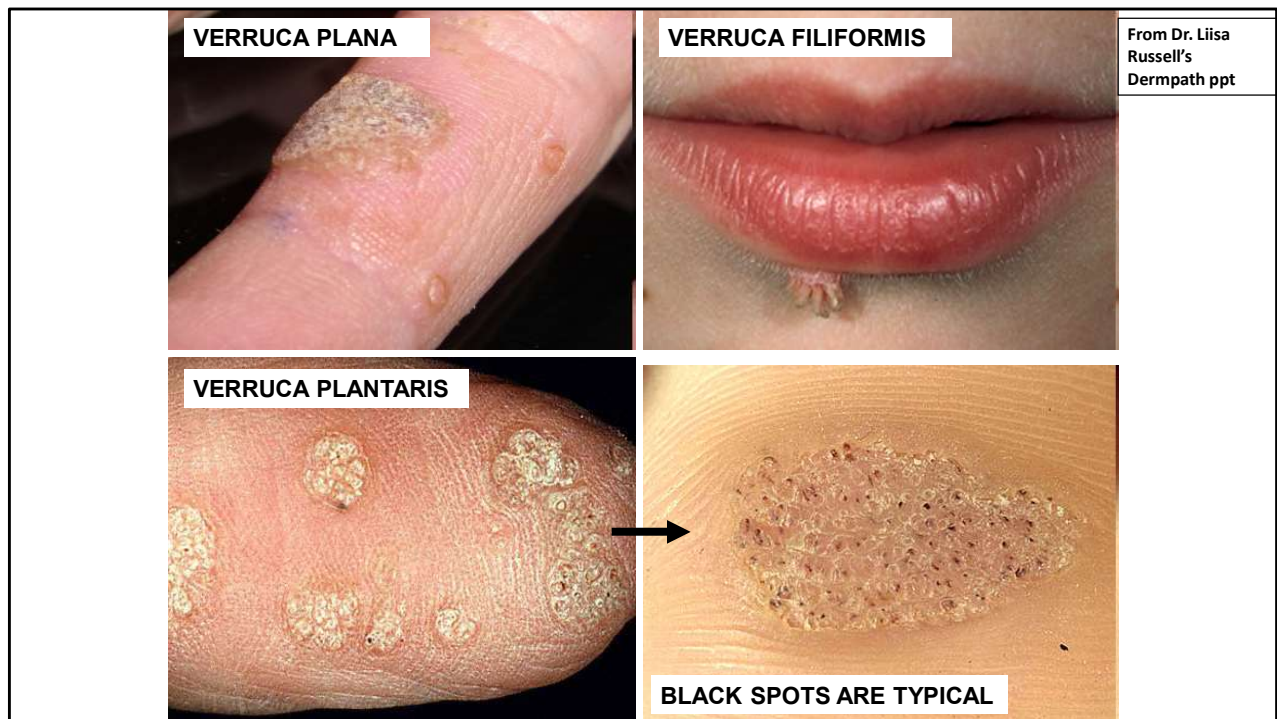
16:18 assoc with cervical cancer



14 yom. Diagnosis: Verruca vulgaris. **Description:** Prominent hyperkeratosis (thickening of the horn layer), acanthosis (thickening of the stratum spinosum), hypergranulosis (thickening of stratum granulosum) and papillomatosis (lengthening of the rete ridges). The elongated rete ridges are directed towards the center of the lesion. Virus particles are mostly found in the zone of hypergranulosis between two papillae. **Clinical findings:** Circumscribed skin-colored papules on the back of the hand. **Stain:** H&E.

HPV 6 and 11 usually adults only

see veruca in a child usually due to abuse



spots correspond to hemorrhage



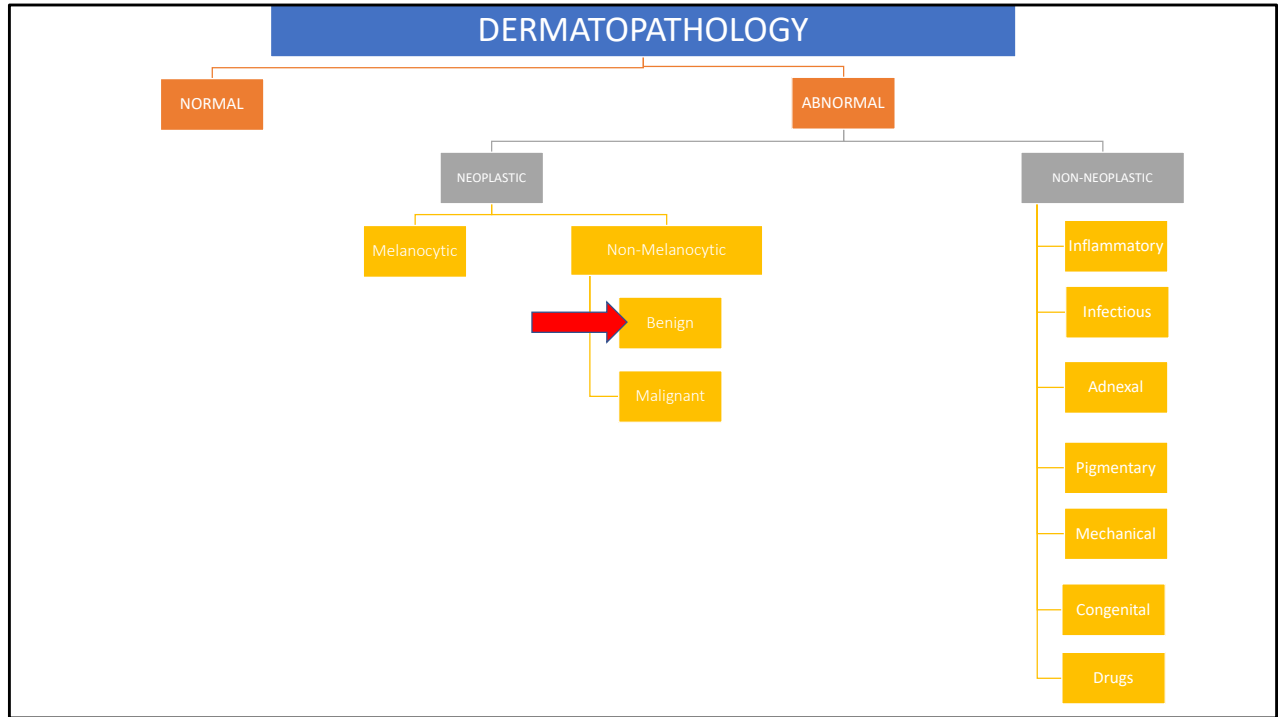
CONDYLOMA ACUMINATUM

From Dr. Liisa
Russell's
DermPath ppt

- HPV types 6 and 11 are responsible for 90% of documented cases of genital warts.

HPV AND CANCER

- HPV 16 and 18 cause the most cervical cancers (70%); 58, 33, 45, 31, 52, 35, 39, 59 are also implicated, but less so.
- HPV 16 and 18 also linked with cancers of the vulva, vagina, penis, anus, and oropharynx
- A relationship between several types of HPV and BCC and SCC have been suggested
- Immunocompromised patients have a higher cancer risk
- HPV has been identified in actinic keratosis
- HPV may play a role in the induction but not in the maintenance of cutaneous SCC (next generation sequencing has produced conflicting results)



FIBROEPITHELIAL POLYP AKA SKIN TAG

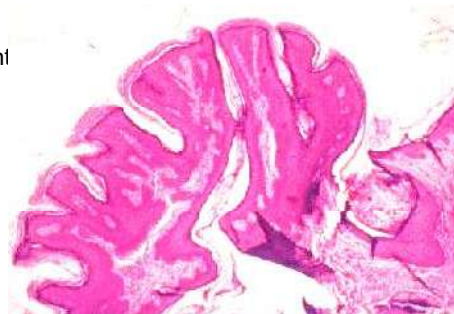
From Dr. Liisa
Russell's
Dermopath ppt

CLINICAL FEATURES:

- Flesh-colored, pedunculated tags of skin in the neck, axilla, groin, or under a woman's breast
- Extremely common

PATHOPHYSIOLOGY IN BRIEF:

- Benign
- Usually inconsequential, but if multiple, may be associated with obesity, diabetes, and intestinal polyposis
- Often become more numerous or prominent during pregnancy, presumably related to hormonal stimulation
- Torsion may cause ischemic necrosis and precipitate removal



Skin tags are extremely common, flesh-colored, pedunculated tags of skin that usually arise in flexural areas (the neck, axilla groin, or under a woman's breast
RIGHT: Multiple skin tags of the axilla. Patients often want these tags removed because they get irritated and may bleed.

Epithelial/follicular inclusion cyst

Most common cutaneous cysts

Etiology and Pathology:

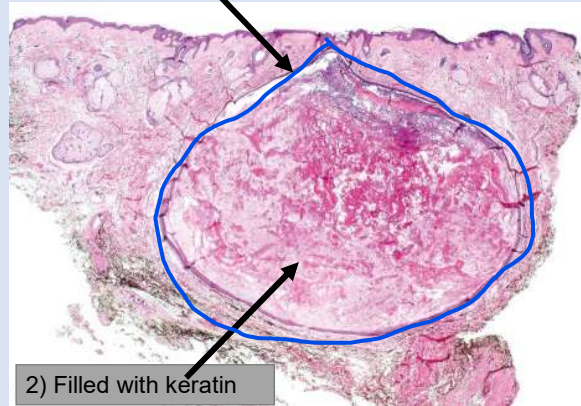
- Derive from follicle or implantation of the epidermis (i.e., sewing needle penetrates skin, resulting in squamous epi being implanted into dermis)
- Multiple cysts can be seen in Gardner's syndrome

Clinical:

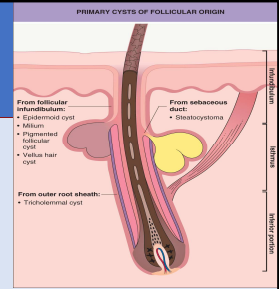
- Solitary, head & neck, 1-4 cm in diameter
- Smooth dome-shaped swellings with punctum
- Can get infected; malodorous 'cheesy' material if you cut these open



1) Cystic invagination **lined by stratified squamous epithelium**



2) Filled with keratin



smell bad bc trapped for a long pd of time

Cutaneous cysts

Calonje, Eduardo, MD, DipRCPPath, McKee's Pathology of the Skin, Chapter 34, 1680-1697.e8

Epidermoid cyst: (A) in this example, the punctum is present; (B) the cyst wall is composed of squamous epithelium and includes a granular cell layer. Note the laminated keratin.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

Cysts

Stone, Mary Seabury, Dermatology, 110, 1917-1929

Primary cysts of follicular origin. Anatomic origin of cysts derived from the pilosebaceous unit. Adapted from Requena L, Sanchez Yus E. Follicular hybrid

cysts. An expanded spectrum. Am J Dermatopathol. 1991;13:228–33.

Copyright © 2018 © 2018, Elsevier Limited. All rights reserved.

Cutaneous cysts

Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 34, 1680-1697.e8

Epidermoid cyst: a typical dome-shaped swelling with two puncta. By courtesy of R.A. Marsden, MD, St George's Hospital, London, UK.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

Cysts

Stone, Mary Seabury, Dermatology, 110, 1917-1929

Epidermoid cyst. Typical clinical appearance of an epidermoid cyst with a yellowish hue. Two pores are present in this example.

Copyright © 2018 © 2018, Elsevier Limited. All rights reserved.

Hemangioma

Cherry angiomas are the most common acquired cutaneous vascular proliferation

Etiology and Pathology:

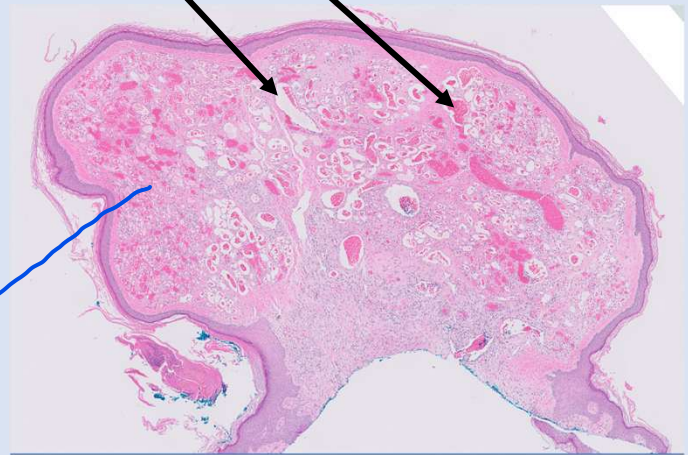
- Overarching term for a benign proliferation of blood vessels
 - Cherry angioma, lobular capillary hemangioma (i.e., pyogenic granuloma), etc...

Clinical:

- Most people >60 years old have one or multiple of these bright red papules
 - Mainly on the trunk



Multiple dilated and congested capillaries in the papillary dermis



too many blood vessels for normal skin

Vascular Neoplasms and Neoplastic-Like Proliferations. North, Paula E., Dermatology, 114, 2020-2049. Cherry angioma. Multiple compressible red papules. Courtesy, Jean L Bologna, MD. Copyright © 2018 © 2018, Elsevier Limited. All rights reserved.

Seborrheic keratosis

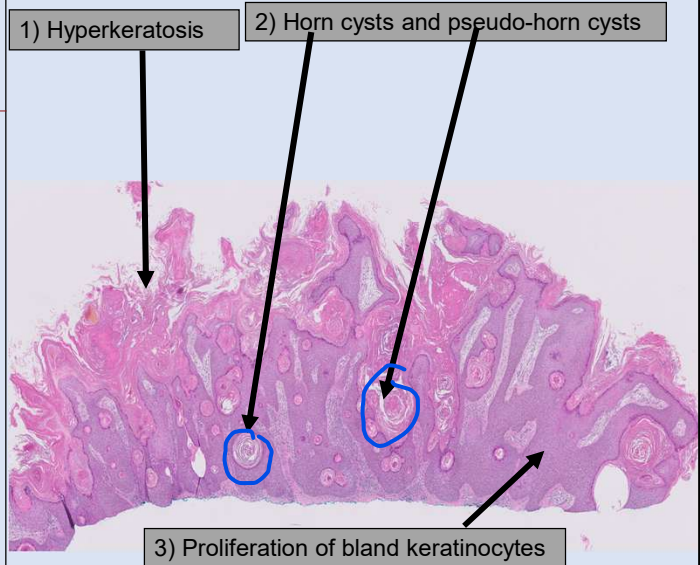
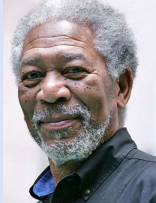
Most common benign skin tumor that look "stuck-on" the skin

Etiology and Pathology:

- Fibroblast growth factor receptor 3 (FGFR3) mutations result in increased keratinocyte proliferation

Clinical:

- Keratotic, exophytic, plaques with tan-to-dark brown cobblestone surface
- Leser-Trélat sign: sudden onset of numerous SKs that may indicate internal malignancy (gastric adenocarcinoma)**
- Dermatosis papulosa nigra: multiple SKs on the face present in up to 35% of African American adults



hyperkeratosis

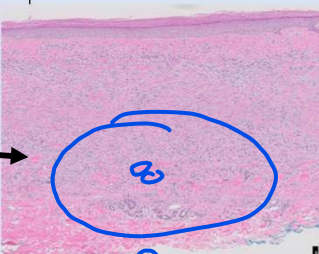
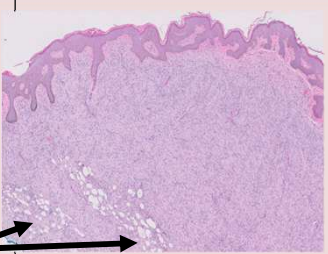
horn cysts

Clinical images of SKs courtesy of Dr. Liisa Russell.

akanthotic

sarcoma - soft tissue malignancy that can metastasize

look like monotonous cells that cartwheel around each other

Dermatofibroma <i>Most common mesenchymal neoplasm in the skin; benign</i>	Dermatofibrosarcoma protuberans <i>Intermediate tumor, can undergo fibrosarcomatous transformation--> metastasize</i>
<u>Etiology and Pathology:</u> Dermal proliferation of spindled cells with "collagen trapping" 	<u>Etiology and Pathology:</u> - COL1A1:PDGFB translocation in majority - Dermal proliferation of <u>uniform</u> spindle cells with storiform (cartwheeling) pattern Infiltrates subcutaneous adipose in a "honeycomb" pattern 
<u>Clinical:</u> - Commonly in 3rd-4th decades, females, legs - Isolated dome-shaped papule or nodule - Sometimes occur after trauma or insect bite - Usually no treatment required 	<u>Clinical:</u> - Indurated, slow-growing plaque on trunk in young-to-middle aged adults 20-50% recur but 5 year survival ~ 99% - Fibrosarcomatous DFSP has metastatic potential - Complete resection with negative margins 

if pinch lesion will dimple = dermatofibroma

indurated = skin component, worrisome
invasive squamous cell carcinoma
if melanocyte = melanoma

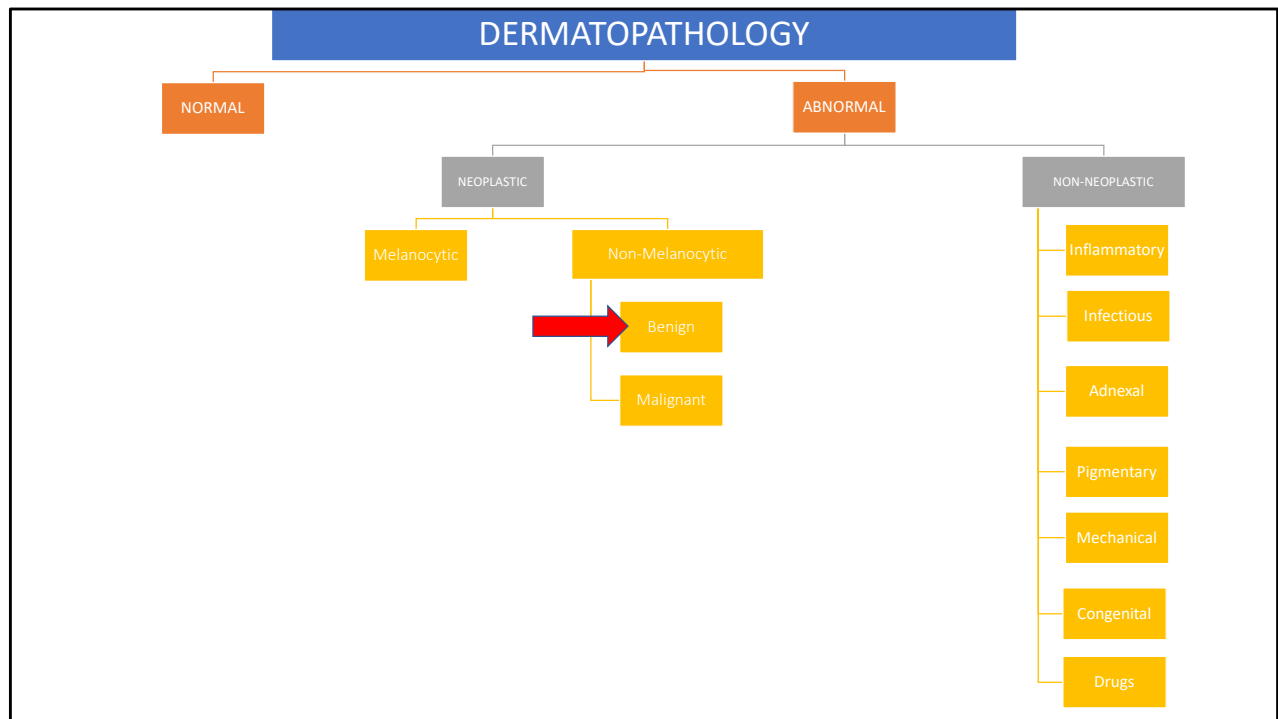
Uptodate: This firm, hyperpigmented papule on the shin is a dermatofibroma.
Originally from visualdx.com.

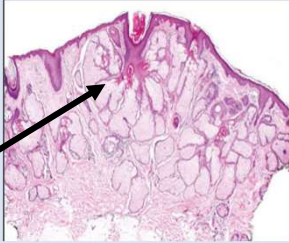
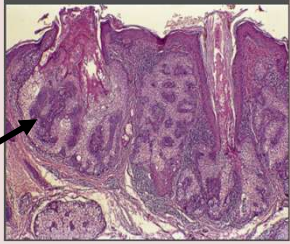
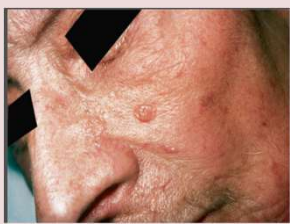
Connective tissue tumors

Calonje, Eduardo, McKee's Pathology of the Skin, Chapter 35, 1698-1894.e88

Dermatofibrosarcoma protuberans: close-up view. By courtesy of M.H.A. Rustin, MD, Royal Free Hospital, London, UK.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.



Sebaceous hyperplasia <i>Common; yellow papules on face of older adults; common</i>		Sebaceous adenoma Rare; well-circumscribed yellow-tan nodule on head and neck in adults. Some are seen in the setting of Muir-Torre syndrome.	
<u>Etiology and Pathology:</u> - Since they do not regress, it is unclear if they truly represent hyperplasia - Hyperplastic sebaceous glands that are present higher in the dermis than normal that usually drain into a central duct		<u>Etiology and Pathology:</u> <i>Some are seen in the setting of Muir-Torre syndrome which is a phenotypic variant of Lynch syndrome (hereditary nonpolyposis colorectal carcinoma syndrome)</i> - Pathology: Lobules of mature sebocytes + rim of immature basaloid cells	
<u>Clinical:</u> Yellowish, dome-shaped papules 1-2 mm in diameter, umbilicated		<u>Clinical:</u> - Well-circumscribed yellow-tan nodule on the head and neck in adults > 40 years old; Muir-Torre syndrome associated lesions can be multifocal - Simple excision is curative; however, do colorectal carcinoma screening in those that are due to Muir-Torre syndrome	

blue/purple = more stem cells benign unless assoc with muir-torre

McKee Pathology of the skin:

Fig. 32.3

Sebaceous hyperplasia: scanning view showing sebaceous glands grouped around a cystic infundibulum.

Fig. 32.1

Sebaceous hyperplasia: multiple lesions are present on the cheek. Note the central umbilication.

By courtesy of the Institute of Dermatology, London, UK.

Fig. 32.22

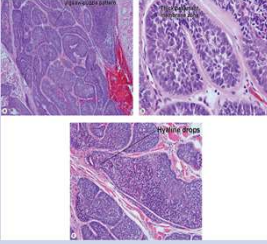
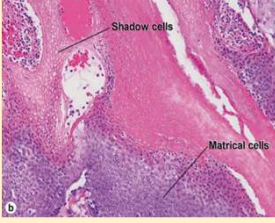
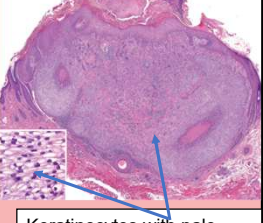
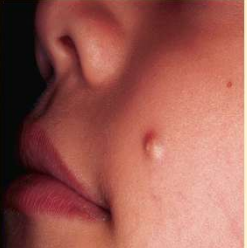
Sebaceous adenoma: note the yellow papule on the cheek of this elderly patient.

By courtesy of R.A. Marsden, MD, St George's Hospital, London, UK.

not basal cell carcinoma no
palisading retraction

sweat



Cylindroma <i>Ductal differentiation, forehead, "turban tumor"</i> Etiology and Pathology • Brooke-Spiegler syndrome • "Jigsaw puzzle" of basaloid cells 	Pilomatixoma <i>Hair follicle differentiation, most common non-melanocytic skin tumor excised in kids</i> Etiology and Pathology: • Beta-catenin mutations in most (CTNNB1 gene) • Basaloid cells + shadow (ghost) cells 	Trichilemmoma <i>Hair follicle differentiation, multiple are major dx criterion for Cowden syndrome</i> Etiology and Pathology: • Multiple in Cowden syndrome (part of PTEN hamartoma tumor syndrome; increased risk of breast, thyroid, endometrial ca) 
Clinical • "Turban tumor" is confluent cylindromas seen in familial cylindromatosis 	Clinical • Commonly on the cheek in kids • Can be firm due to calcification or ossification 	Clinical • For Cowden syndrome patients, earlier cancer surveillance (i.e., women undergo mammography at age 30)  Multiple papules in Cowden syndrome can coalesce on buccal mucosa or tongue and impart "cobblestone" appearance

calcifications in benign tumor

Turban tumor clinical photograph courtesy of Dr. Liisa Russell

((Pilar and sebaceous neoplasms Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 4, 68-82 Pilomatricoma Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.))

(Sweat gland neoplasms

Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 5, 83-100

Cylindroma

Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.

Adnexal Neoplasms

McCalmont, Timothy H., Dermatology, 111, 1930-1953

Tricholemmoma. Circumscribed nodular proliferation of pale-staining cells with a broad connection to the overlying epidermis. There is also surface crust and focal areas of necrosis within the tumor. Sometimes these lesions are misdiagnosed as sup...

Copyright © 2018 © 2018, Elsevier Limited. All rights reserved.

Tumors of the hair follicle

Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 31, 1545-1588.e16

Cowden disease: the innumerable shiny papules on this patient's tongue are characteristic. By courtesy of R. Graham, MD, and R. Emmerson, MD, Royal Berkshire Hospital, Reading, UK.

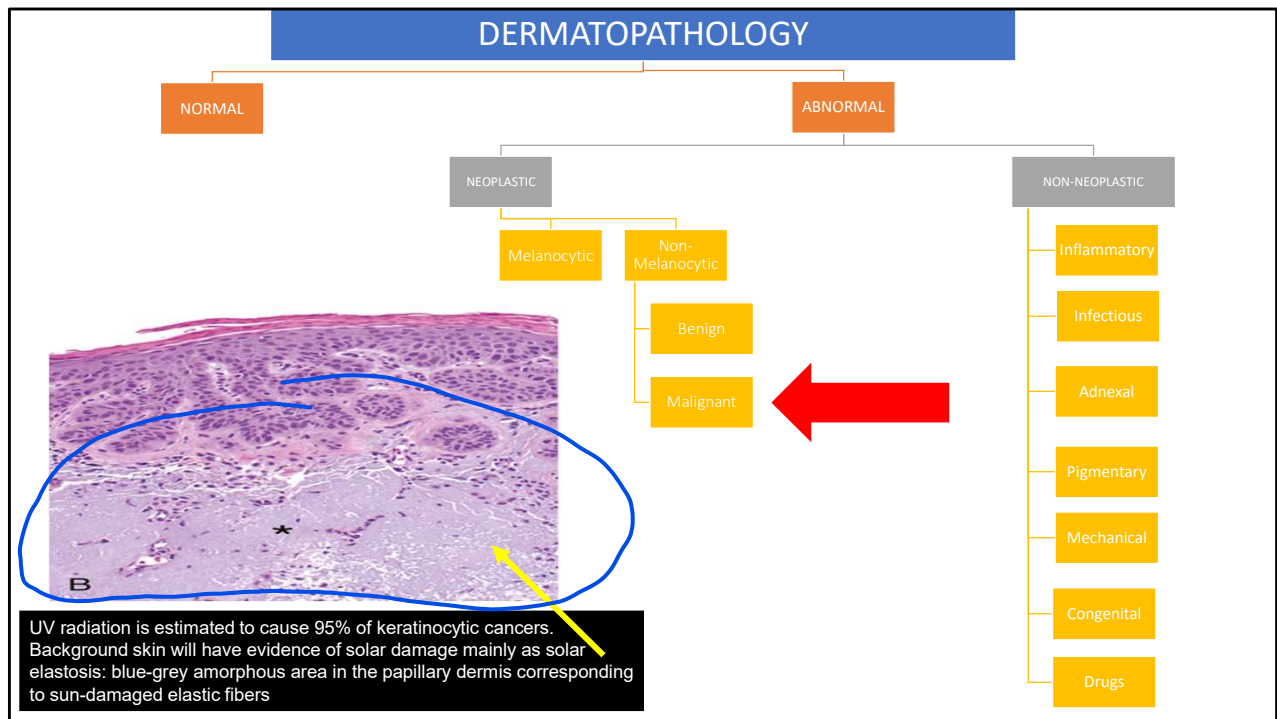
Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

Adnexal Neoplasms

McCalmont, Timothy H., Dermatology, 111, 1930-1953

Pilomatricoma. A firm nodule on the cheek of a child.

Copyright © 2018 © 2018, Elsevier Limited. All rights reserved.



"liquid dermis"

- Skin
Kumar, Vinay, MBBS, MD, FRCPath, Robbins & Kumar Basic Pathology, 22, 775-793

Actinic keratosis.

Copyright © 2023 Copyright © 2023 by Elsevier Inc. All rights reserved.

squamoid = more pink

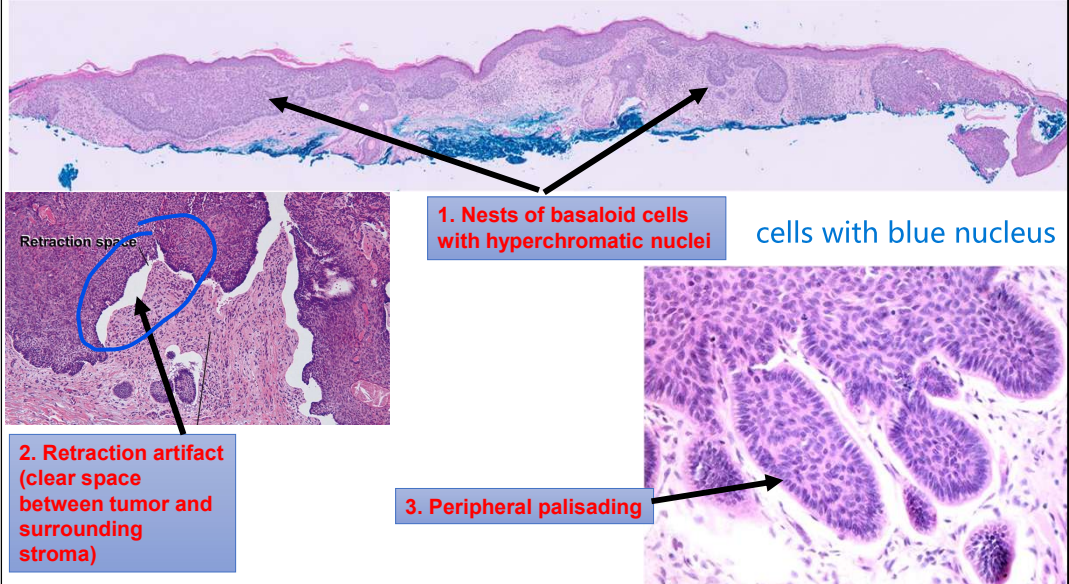


Basal cell carcinoma

Most common malignancy in White populations. Nests of basaloid cells with "picket fence" palisading

Etiology and Pathology:

- Arise from basal cells of epidermis or hair follicle stem cells
- **UV radiation** (causes mutations in PTCH1 and TP53)
- Patients who are older, male, have light skin pigmentation and **exposed to intense intermittent UV radiation** are most likely to develop BCC



Photomicrograph of "3. Peripheral palisading" from Dr. Liisa Russell's Dermath ppt

Actinic Keratosis, Basal Cell Carcinoma, and Squamous Cell Carcinoma
Soyer, H. Peter, Dermatology, 108, 1872-1893

Basal cell carcinoma (BCC): range of histopathologic findings. A Superficial type with a few basaloid buds and stromal retraction spaces. B A single basaloid aggregation with all the morphologic findings (e.g. peripheral palisading of nuclei, retraction...

Copyright © 2018 © 2018, Elsevier Limited. All rights reserved.

Malignant tumors of the epidermis. Dirk M. Elston. Dermatopathology, Chapter 3

Basal cell carcinoma

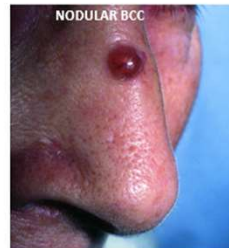
Most common malignancy in White populations. Nests of basaloid cells with "picket fence" palisading at base

Clinical

- Pearly nodules with telangiectasias and rolled borders
- Head and neck > trunk and extremities
- Very rarely metastasize.
- Nevoid basal cell carcinoma syndrome (Gorlin syndrome): rare, auto dominant, multiple BCCs beginning in childhood



GALLERY OF BASAL CELL CARCINOMAS



From Dr. Liisa Russell's Dermopath ppt

can be locally aggressive

if seeing in a younger pt think gorlin syndr

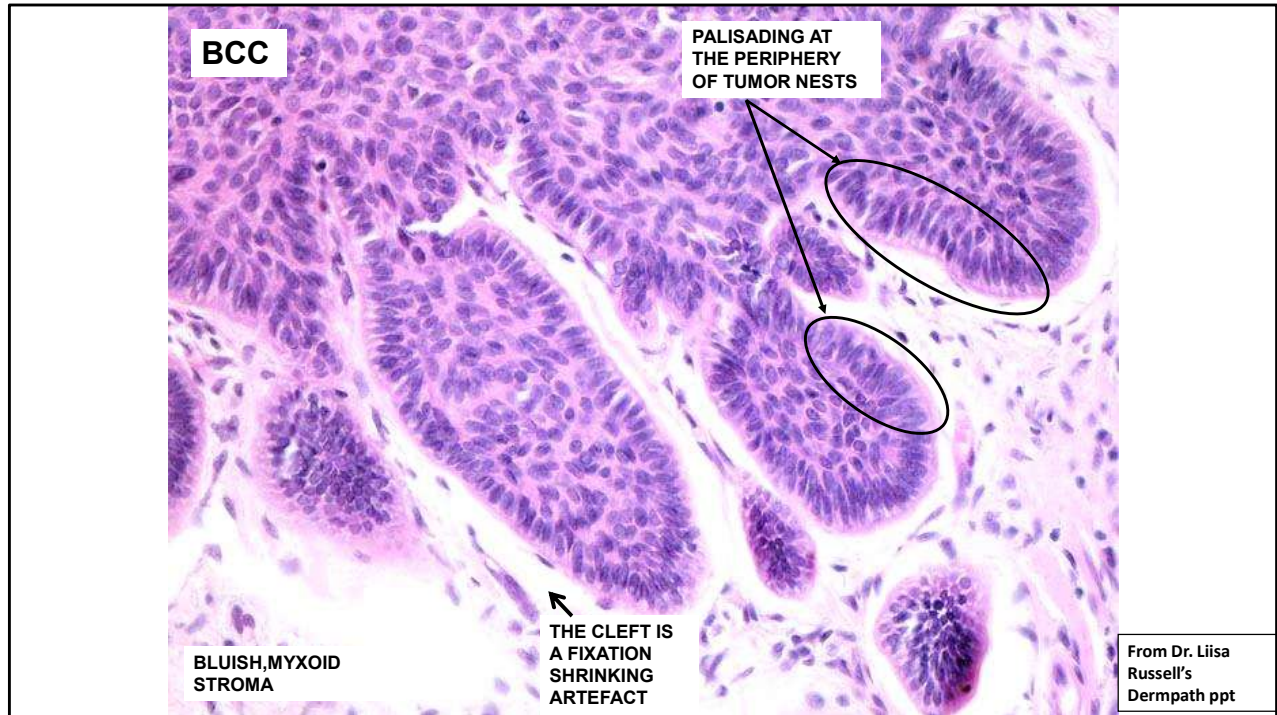
Tumors of the surface epithelium

Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 24, 1156-1233.e39

Ulcerative basal cell carcinoma: note the central ulceration and characteristic rolled border. By courtesy of R.A. Marsden, MD, St George's Hospital, London, UK.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

has melanocytes throughout



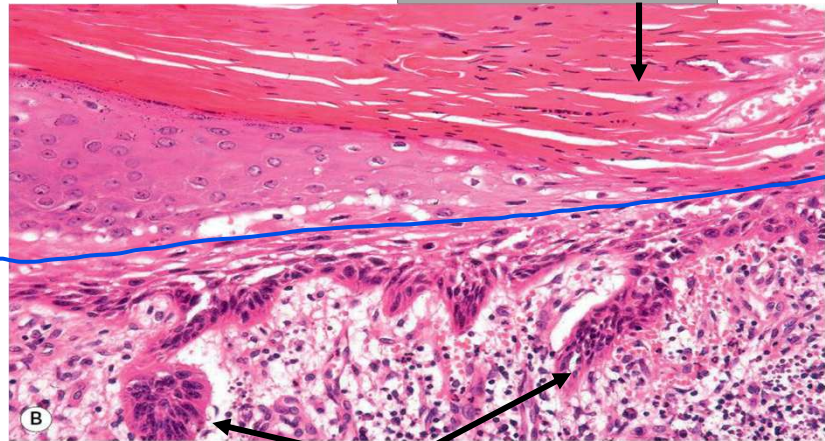
Diagnosis: Basal cell carcinoma. **Description:** Solid basaloid tumor cells. Palisading. Artificial cleft.

Actinic keratosis

"Pre-cancer" with rate of progression to squamous cell carcinoma from 0.1 to 2.6% per year. Squamous dysplasia does NOT affect the full-thickness epidermis.

Etiology and Pathology:

- **UV-induced DNA damage** associated with TP53 mutations and other genes frequently mutated in SCC
 - Since AKs are on a continuum with SCCs



1) Squamous dysplasia (i.e., enlarged, hyperchromatic nuclei, mitotic figures) that is not full thickness

abnormal retention of nuclei

if full thickness squamous cell carcinoma in situ

Actinic keratosis

"Pre-cancer" with rate of progression to squamous cell carcinoma from 0.1 to 2.6% per year.

Clinical:

- Scaly, erythematous macules or papules at sites of chronic sun exposure (scalp, face, lateral neck, dorsal hands and forearms, ears, lip (actinic cheilitis))
- Usually don't need to biopsy since visual and tactile diagnosis
 - Unless you need to distinguish from SCC (i.e., lesions that are >1 cm, indurated, ulcerated, rapidly growing, fail to respond to appropriate therapy)
- **While majority don't progress, 60% of cutaneous SCCs arise in sites of pre-existing actinic keratoses**



ACTINIC KERATOSIS

From Dr. Liisa Russell's
Dermopath ppt



Extreme hyperkeratosis in actinic keratosis may cause development of a cutaneous horn

Tumors of the surface epithelium

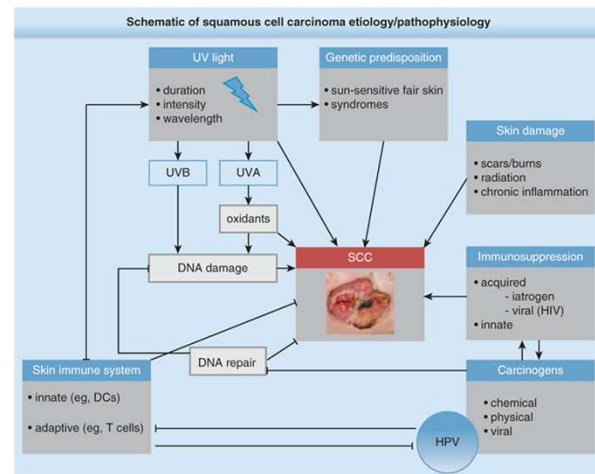
Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 24

Squamous cell carcinoma

2nd most common skin cancer. Called "In situ" if full-thickness atypia vs. "invasive" if nests infiltrate into the dermis.

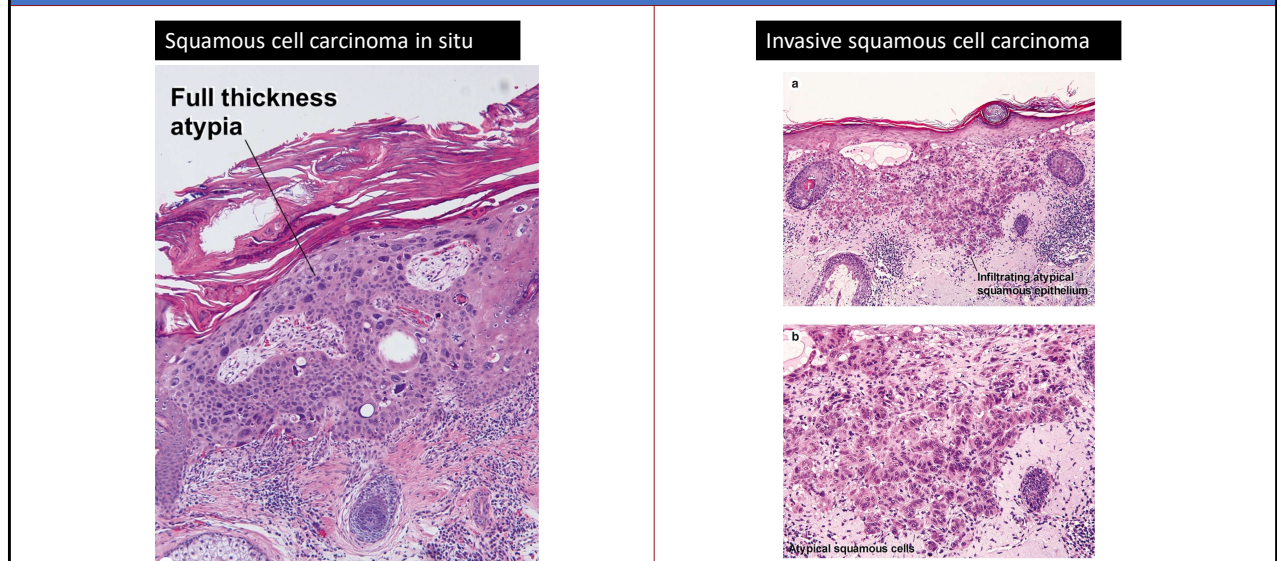
Etiology and Pathology:

- **UV radiation** causes widespread DNA damage. Other risk factors: immunosuppression, oncogenic HPV (genital skin, nail), arsenic (drinking water from wells), chronically inflamed skin
 - **SCC occurring in chronic wound is known as Marjolin's ulcer** (more aggressive course)
- SCCIS and AK are precursors to invasive SCC



Squamous cell carcinoma

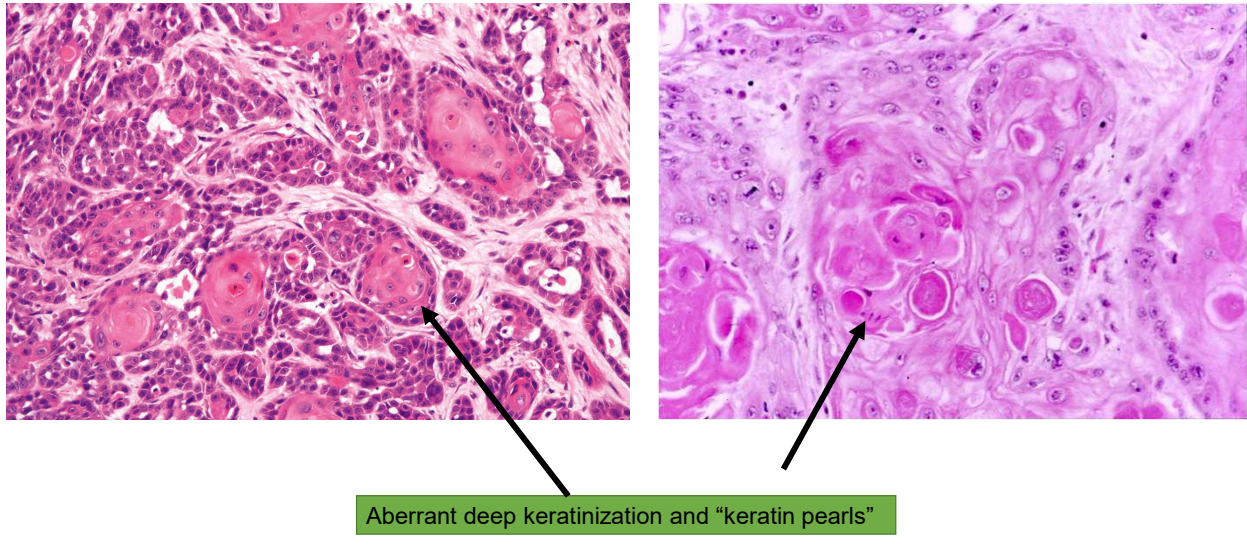
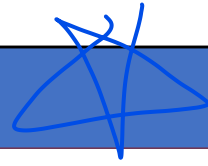
2nd most common skin cancer. Called "In situ" if full-thickness atypia vs. "invasive" if nests infiltrate into the dermis.



big cells that transverse but not invading dermis

Malignant tumors of the epidermis. Dirk M. Elston. Dermatopathology, Chapter 3

Invasive squamous cell carcinoma



malignant keratinocytes producing when not supposed to

Malignant tumors of the epidermis

Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 3, 54-67

Bowen disease

Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.

Tumors of the surface epithelium

Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 24, 1156-1233.e39

Well-differentiated squamous cell carcinoma: scanning view showing an obviously keratinizing tumor infiltrating deep into the reticular dermis.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

Tumors of the surface epithelium

Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 24, 1156-1233.e39

Moderately differentiated squamous cell carcinoma: there is much more obvious pleomorphism. Nevertheless, the squamous nature of the tumors is still readily apparent.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

Squamous cell carcinoma

2nd most common skin cancer. Called "In situ" if full-thickness atypia vs. "invasive" if nests infiltrate into the dermis.

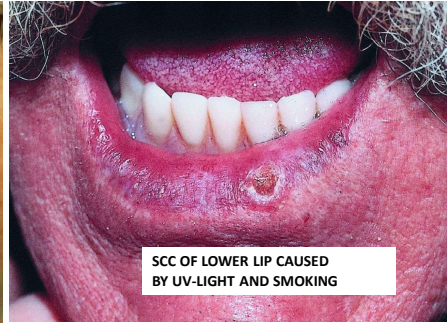
Clinical:

- **SCCIS:**
 - Solitary, scaly erythematous patch or plaque that can usually remain unchanged for years on sun-exposed skin
 - 4% of SCCIS lesions progress to invasive SCC
- Invasive SCC:
 - Solitary red scaly plaque progressing to hyperkeratotic, ulcerated nodule
 - Chronically sun-exposed skin of head and neck > trunk and legs
 - Overall 5-year cure rate > 90%
 - 2-5% metastasize
 - **Patients with xeroderma pigmentosum, vitiligo, and albinism are at increased risk of SCC**
 - High-risk histologic prognostic features are tumor thickness >2 mm, clark level IV or V invasion, perineural invasion, primary site on ear or lip, and poor differentiation

once metastasize very hard to treat

like to ask about XP being high risk

SQUAMOUS CELL CARCINOMA (SCC)



CLINICAL FEATURES:

- Indurated, scaling, erythematous papules, nodules or plaques that occasionally ulcerate and bleed.
- Often develops in a preexisting actinic keratosis.
- The second most common skin cancer.



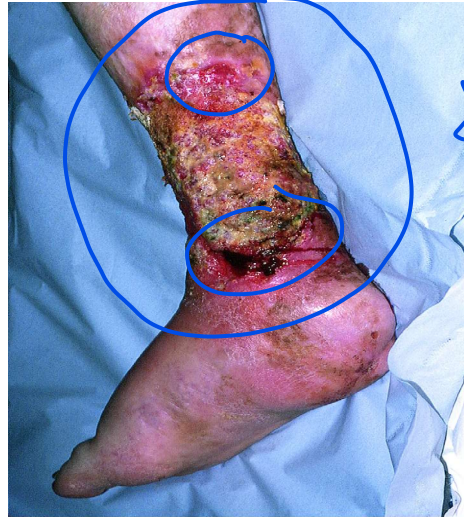
From Dr. Liisa Russell's
Dermopath ppt

most frequently on lower lip

**CHRONIC RADIODERMATITIS
WITH DEVELOPING SCC**



**CHRONIC STASIS ULCER CAN
LEAD TO SCC** chronic inflammation



From Dr. Liisa Russell's
Dermopath ppt

large ulcerate plaque with overlying scale and crust probably associated with bacteria

Keratoacanthoma

Considered a well-differentiated subtype of SCC that looks like "keratin plugged volcano," rapidly grows, and can spontaneously regress.

Etiology:

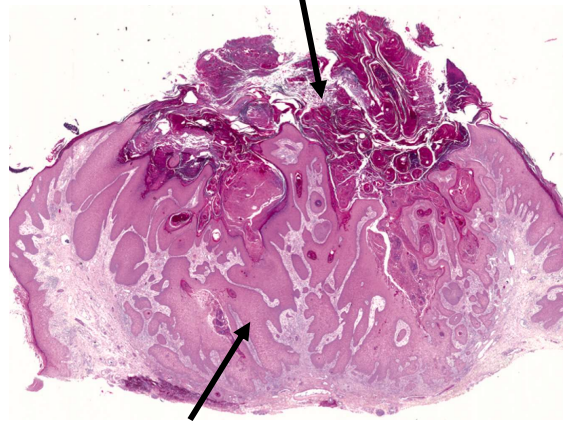
- **UV radiation**

Clinical:

- "Crateriform" nodule with a central depression filled with keratin



1) Central keratin filled crater



2) Glassy pink keratinocytes that are well-differentiated

raised nodule with prominent keratotic center

Malignant tumors of the epidermis

Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 3, 54-67

Keratoacanthoma

Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.

FYI: Sebaceous carcinoma

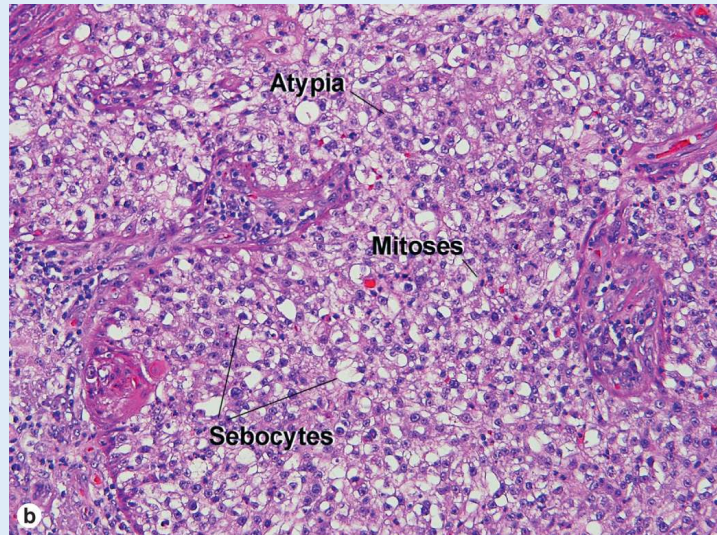
Malignant epithelial neoplasm demonstrating sebaceous differentiation

Etiology:

- Risk factors include UV-exposure, prior radiation, Muir-Torre syndrome

Clinical:

- Head and neck (periocular eyelid in 75%)
- 10-30% risk of tumor-related mortality



sebaceous - mire-torre, hereditary colon cancer

Tumors and related lesions of the sebaceous glands

Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 32, 1589-1610.e11

cardinal signs: neovascular invasion, atypia, irregular nuclear contours, mitotic figures

Pilar and sebaceous neoplasms

Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 4, 68-82

Sebaceous carcinoma

Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.

Sebaceous carcinoma: (A) an encrusted tumor on the scalp of an elderly patient; By courtesy of D.H. McGibbon, St Thomas' Hospital, London, UK.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

chronic

Mycosis fungoides & Sézary syndrome

Neoplastic T-cells that home to the skin (cutaneous T-cell lymphoma) and look "cerebriform"

Etiology:

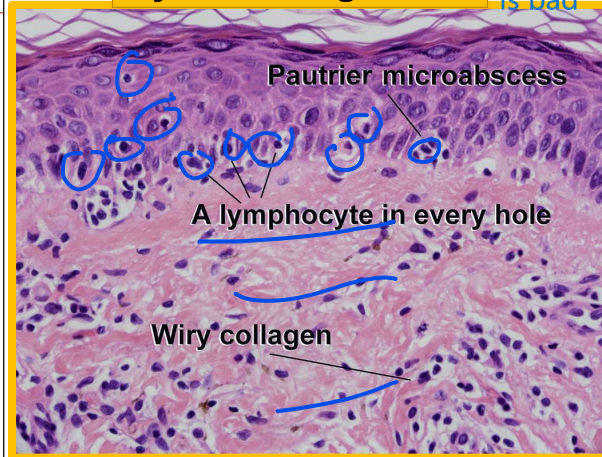
- Unknown

Clinical:

- Erythrodermic rash that can have various phases (plaque, tumor)
- **Indolent course over decades**
- **Excellent prognosis, if limited disease**
- Sézary syndrome is erythroderma + lymphadenopathy + **cerebriform** clonal T cells in skin/lymph node/peripheral blood

Mycosis fungoides

lining up
is bad



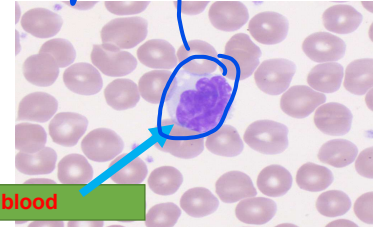
fibrosis

Atypical "cerebriform" lymphocyte in peripheral blood

Sézary Syndrome



Electron micrograph of Sézary cell showing nuclei with invaginated nuclear membranes and deep nuclear clefting.



B cell lymphoma - CD20 respond to rituximab, t cells don't have equivalent except for mycosis fungoides large cell transformation - main cause of morbidity

- *WHO 5th edition Derm*

Mycosis fungoides: large erythematous plaques with scaling.



unless long time, not working with therapy, bathing suit distribution

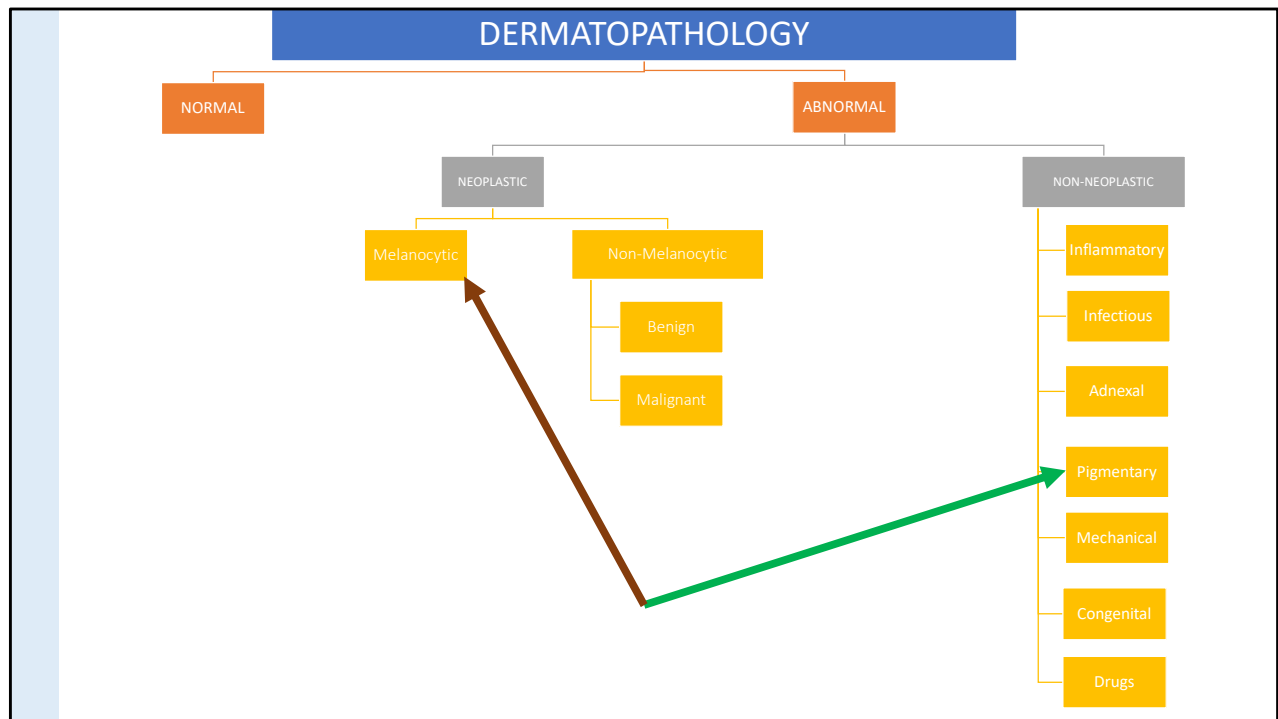
Cutaneous lymphoproliferative diseases and related disorders

Goodlad, John, McKee's Pathology of the Skin, Chapter 29, 1403-1519.e45

Mycosis fungoides: large erythematous plaques with scaling. By courtesy of the Institute of Dermatology, London, UK.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

mycosis fungoides t cell lymph usually indolent can have large cell transform
sezary occurs in skin but also needs to occur in another place like in peripheral



Disorders of Pigmentation and Melanocytes

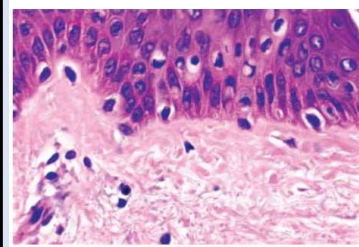
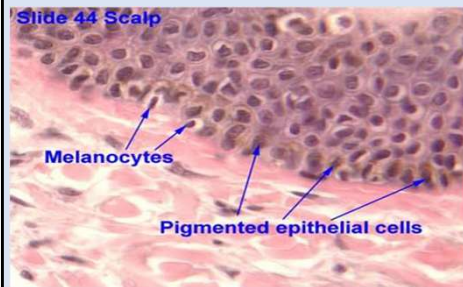
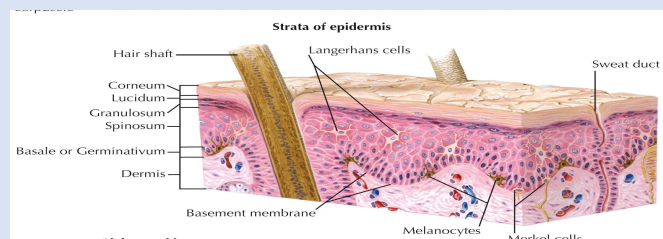
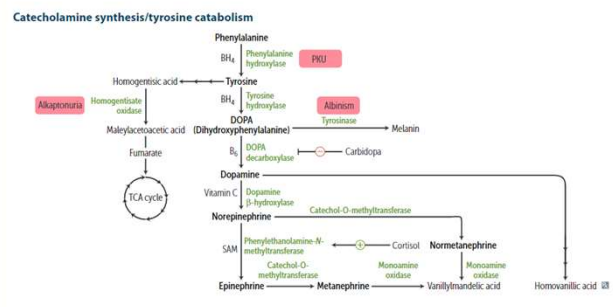


FIGURE 1.8 Melanocytes in the basal layer, composed of ovoid nuclei within a clear space.



FYI:



keratinocytes have pigment in them melanocytes produce pigment

Anatomy, Physiology, and Embryology

Anderson, Bryan E., MD, Netter Collection of Medical Illustrations:
Integumentary System, Section 1, 1-11

(no summary available)

Copyright © 2012 Copyright © 2012 by Saunders, an imprint of Elsevier Inc.

First Aid for the USMLE Step 1 (2022)

Approach to melanocytic lesions

Clinical



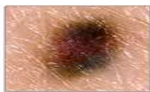
A

is for **ASYMMETRY**
If you draw a line down the middle of the mole, you will find that each half of a melanoma doesn't match in size.



B

is for **BORDER**
The edges of early Melanoma are quite likely to be irregular, crusty or notched.



C

is for **COLOUR**
Healthy moles are uniform in colour. A variety of colours, especially black, white and/or blue is worrying.



D

is for **DIAMETER**
Melanomas are usually larger in diameter than 6 millimetres, although they may be smaller.



E

is for **EVOLVING**
When a mole begins to change in size, shape, colour, or features or develops symptoms like itching, tenderness or bleeding, this points to danger.

Pathologic

- Size
- Symmetry
- Circumscription
- Bridging
- Pagetoid scatter
- Maturation
- Cytology
- Mitoses

Approach to melanocytic proliferations – Adapted from Dr. Rynanne Brown's melanocytic lesions lecture slides (2020)

<https://www.rtwskin.co.uk/melanoma/irregular-beauty-mark-showing-changes-on-skin-in-melanoma-patients/>

Melasma

Facial hyperpigmentation associated with pregnancy or OCP

Etiology and Pathology:

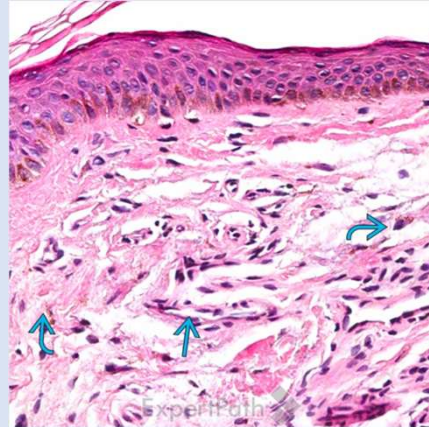
- Unknown (but established association with pregnancy, OCP use)
- Triggered and worsened by sun exposure

Clinical:

- "Mask of pregnancy"



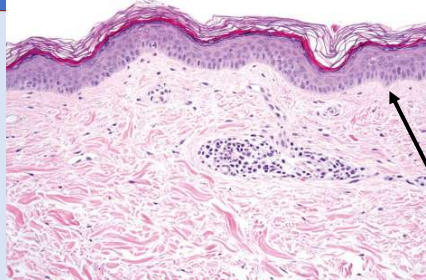
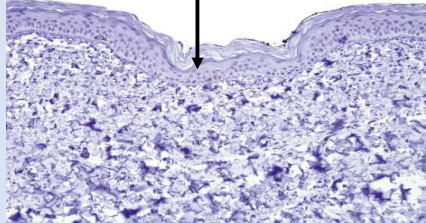
Number of melanocytes is not increased



Increased melanin in the epidermis and dermis (within dermal melanophages)

Clinical photo from Dr. Liisa Russell's Dermopath #2 powerpoint 2022

unfair to show on exam

Vitiligo <i>Autoimmune disorder where destruction of melanocytes causes depigmentation; most frequent cause of skin depigmentation</i>		
<u>Etiology and Pathology:</u> - Unknown (one theory suggests autoimmune destruction of melanocytes) - Associated with other autoimmune diseases (Hashimoto thyroiditis, pernicious anemia, Addison disease, alopecia areata)	 Complete loss of melanocytes by S100 immunohistochemistry and by H&E	
<u>Clinical:</u> Any age, depigmented macules and patches		

brown cells = melanocytes

Clinical photo from Dr. Liisa Russell's DermPath #2 powerpoint 2022

Disorders of pigmentation

Calonje, Eduardo, MD, DipRCPath, McKee's Pathology of the Skin, Chapter 20, 990-1014.e14

Vitiligo: skin biopsy from lesional skin stained with hematoxylin and eosin. There is complete absence of melanocytes and melanin pigment. There is a light perivascular chronic inflammatory cell infiltrate.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

Ephelides (freckles)

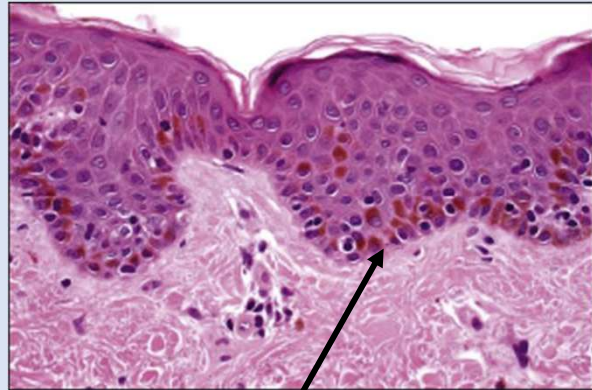
Benign, no propensity for malignant transformation

Etiology and Pathology:

- Hyperpigmentation is thought to be due to increased amounts of melanin pigment within keratinocytes

Clinical:

- Only occur on sun-exposed areas (face)
- More pronounced in summer, fade in winter



Increased basilar pigmentation of keratinocytes
(melanocyte numbers are usually normal)

not fair to show on exam

wax and wane with season

Solar lentigo ("liver spots")

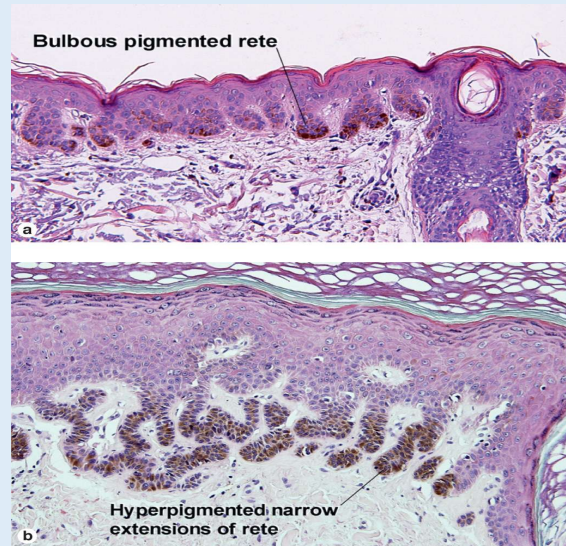
Related to sun exposure, **proliferation of keratinocytes (normal or increased # of melanocytes); doesn't follow ABCDE rules**

Etiology and Pathology:

- Chronic UV-exposure (incidence increased with age)
- Pathology: "dirty socks" hanging down from the epidermis

Clinical:

- Face, forearms, dorsal aspects of hands



Melanocytic neoplasms

Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 6, 101-131

Solar lentigo

Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.

Various theories, but per WHO: "Essential and desirable diagnostic criteria Well-circumscribed intraepidermal keratinocyte proliferation with elongated rete-ridges and lack of cytologic atypia or squamous dysplasia." Vs. Uptodate and Weedon's + others say this is a proliferation of melanocytes.

a nest of melanocytes = mole

Simple lentigo (lentigo simplex)

Not related to sun exposure

Etiology and Pathology:

- Hyperpigmented keratinocytes + variable # of melanocytes, but no nests

Clinical:

- Small, uniformly pigmented
- Large numbers seen in:
 - Peutz-Jeghers
 - Addison disease
 - Carney complex



Increased basilar pigmentation of keratinocytes



Increased number of melanocytes

Macromelanosomes

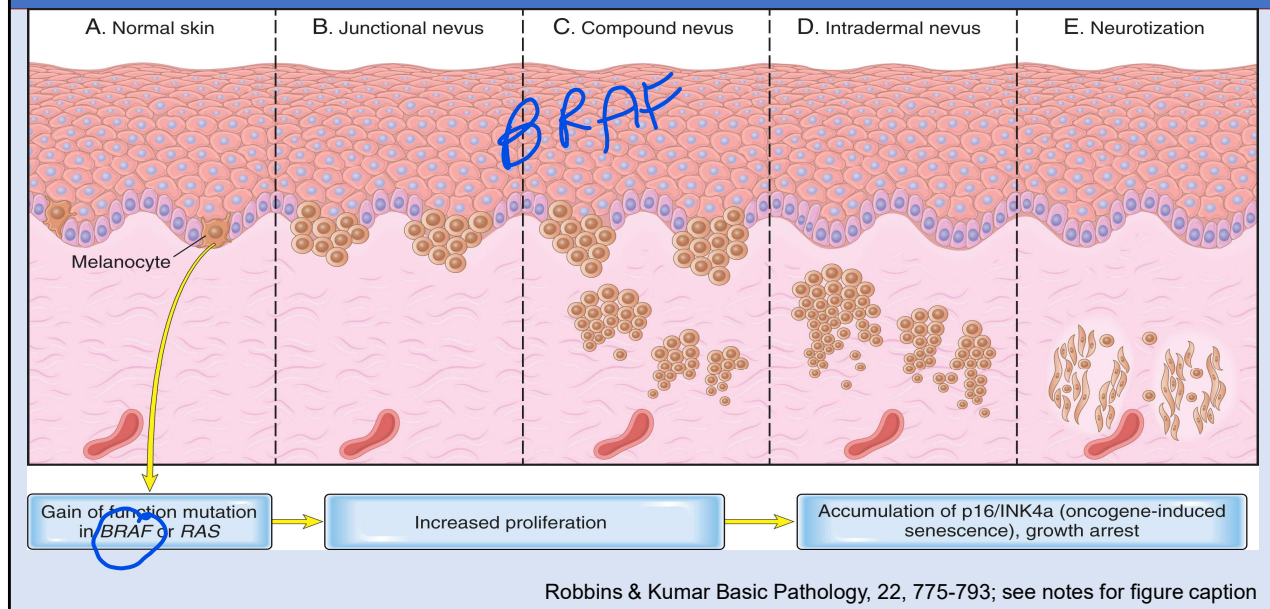
Melanocytic nevi

Luzar, Boštjan, McKee's Pathology of the Skin, Chapter 25, 1234-1309.e16

Lentigo simplex: note the presence of macromelanosomes.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.

Melanocytic Nevi – steps in development



nest = ≥ 3 melanocytes

Skin

Kumar, Vinay, MBBS, MD, FRCPath, Robbins & Kumar Basic Pathology, 22, 775-793

Morphologic and molecular steps in the development of melanocytic nevi. (A) Normal skin uninvolved by nevi shows only scattered melanocytes. (B) An activating mutation in *BRAF* or *RAS* drives proliferation of junctional melanocytes, leading to the formation of a nevus. (C) Over time, nests of melanocytes may penetrate into the dermis, producing a compound nevus. (D, E). Subsequent accumulation of the tumor suppressor molecule p16 (also known as INK4a) appears to induce senescence, leading to permanent growth arrest and “maturation” of intradermal nevoid cells, a process referred to as “neurotization.”

Copyright © 2023 Copyright © 2023 by Elsevier Inc. All rights reserved.

no dermal component

Melanocytic lesions – clinical comparison



Fig. 25.25
Junctional melanocytic nevus: the lesion is small. Note the uniform coloration. By courtesy of M. Liang, MD, The Children's Hospital, Boston, USA.



Fig. 25.176
Dysplastic nevus: nevi are often greater than 6 mm in diameter. Borders are typically irregular. From the collection of the late N.P. Smith, MD, the Institute of Dermatology, London, UK.

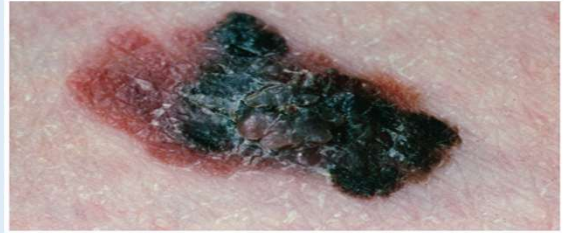


Fig. 26.6
Superficial spreading melanoma: there is a large nodule of invasive melanoma with a surrounding macular component. From the collection of the late N.P. Smith MD, the Institute of Dermatology, London, UK.

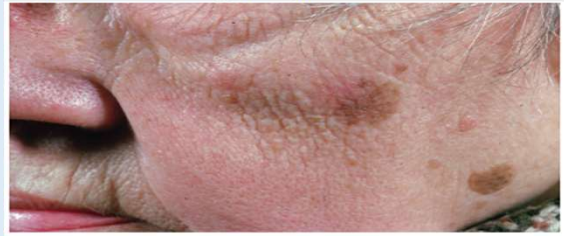


Fig. 26.1
Lentigo maligna: this variably pigmented lesion was present for many years. There is no invasive component. From the collection of the late N.P. Smith MD, the Institute of Dermatology, London, UK.

North, Jeffrey P., McKee's Pathology of the Skin

3 colors = bad sig
evolved over time - pt may not be able to sen

Melanocytic Nevus

Benign proliferation of melanocytes as **nevi**

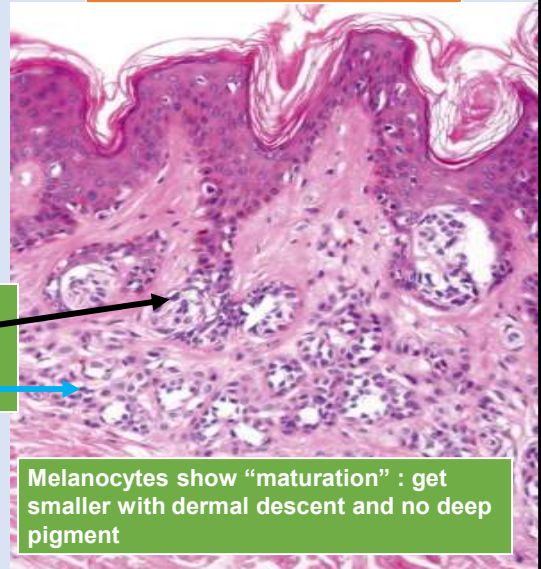
Etiology and Pathology:

- Benign, clonal proliferation of melanocytes
- Must have nests of melanocytes (aggregates of at least 3 melanocytes)
- BRAF mutations
- Symmetric, well-circumscribed, round nests of melanocytes in the epidermis
 - If dermal component is present: melanocytes "mature" or get smaller as they descend into the dermis, with no deep pigment

1. Nests of bland melanocytes

2. Compound = Nests are both **junctional** (in the epidermis) and **within the dermis**

Compound melanocytic nevus



Melanocytic nevi

Luzar, Boštjan, McKee's Pathology of the Skin, Chapter 25, 1234-1309.e16

Compound nevus: both junctional and dermal components are present.

Copyright © 2020 © 2020, Elsevier Limited. All rights reserved.



What is the most likely diagnosis?

- ☒ A. Intradermal melanocytic nevus
- B. Junctional melanocytic nevus
- C. Melanoma

IDN

no junctional component = intradermal

not on exam

INTRADERMAL NEVUS

- DOME-SHAPED
- VARIABLE AMOUNT OF PIGMENT



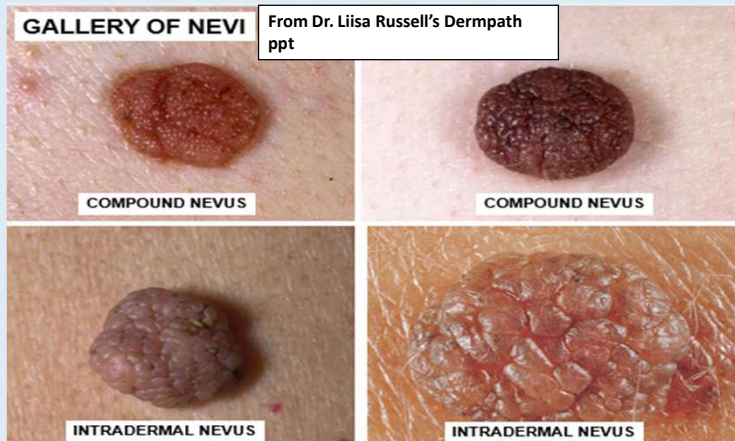
From Dr. Liisa
Russell's
Dermpath ppt

Intradermal nevus

Melanocytic Nevus

Clinical:

- <5 mm, symmetric, well-defined borders, homogenous pigmentation
- Risk of progression to melanoma is very low, but nevi are potential precursors and risk markers
 - Increased # of total or large nevi are at increased risk of melanoma



Spitz nevus

Younger patients (~21 years old); benign with low risk of malignant transformation. "Bananas hanging from the epidermis."

Etiology and Pathology:

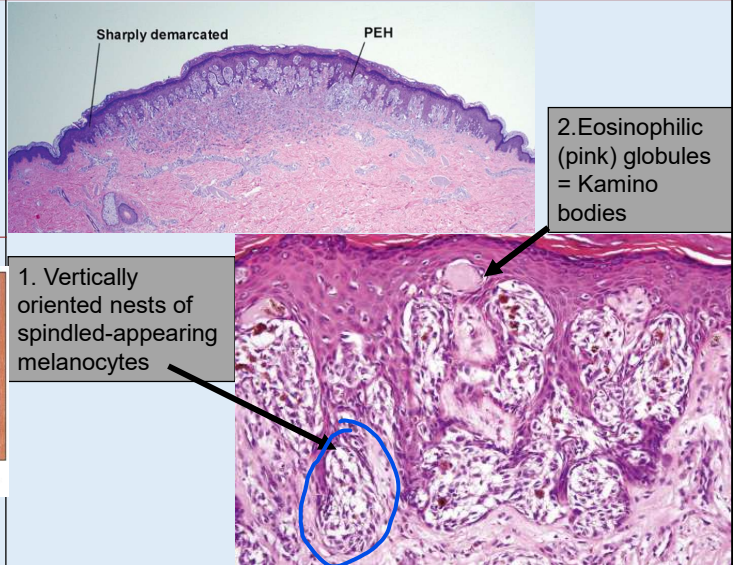
- Usually, HRAS or receptor kinase fusions
- Can appear atypical both clinically and cytologically, but the architectural features (i.e., symmetry, maturation) are reassuring

Clinical:

- Lower extremity, face, trunk
- Typically < 6 mm, can range from red to tan or black



Fig. 25.110
Spitz nevus: the cheek is a commonly affected site. By courtesy of the Institute of Dermatology, London, UK.



*PEH = pseudoepitheliomatous hyperplasia

Melanocytic neoplasms

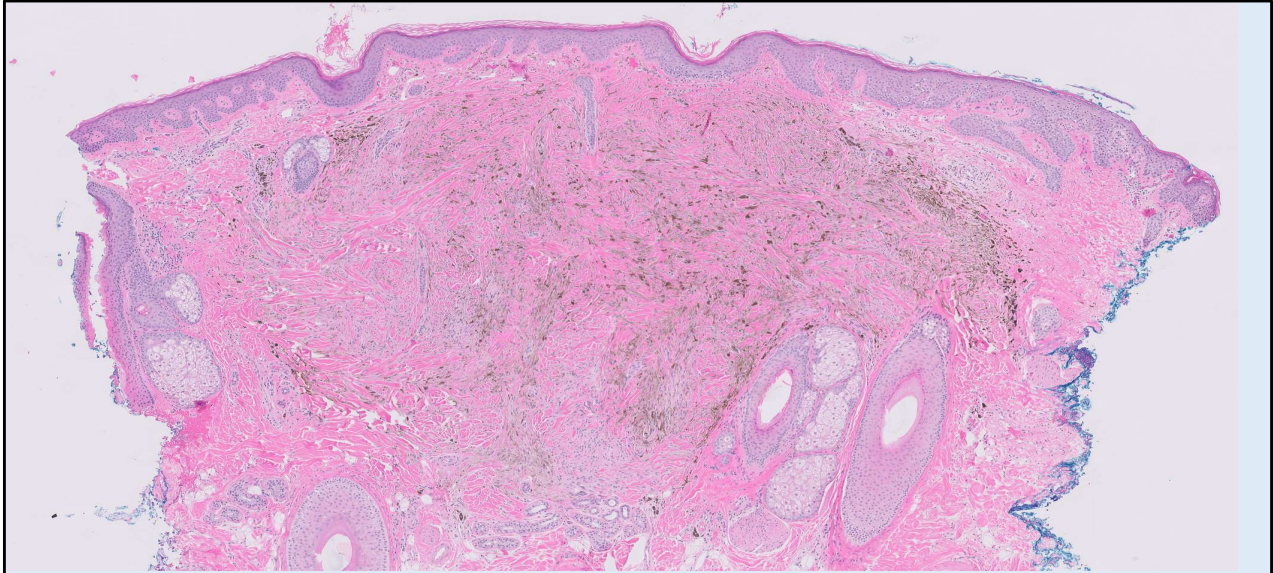
Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 6, 101-131

Spitz nevus (PEH, pseudoepitheliomatous hyperplasia)

Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.

North, Jeffrey P., McKee's Pathology of the Skin

symmetric
blue nevus - pigmented dendritic melanocytes in dermis



What is the most likely diagnosis?

- A. Melanoma
- B. Spitz nevus
- C. Blue nevus

Blue nevus

BLUE NEVUS

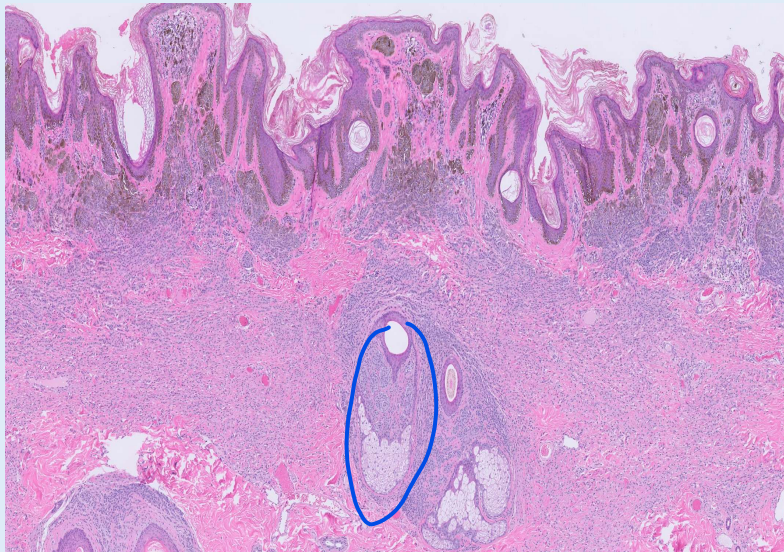
From Dr. Liisa Russell's Dermpath ppt



COMMON BLUE NEVUS:
IF PRESENT FOR A LONG TIME, DOES NOT
NEED TO BE BIOPSIED



CELLULAR BLUE NEVUS:
MUST BE BIOPSIED TO RULE OUT
MELANOMA



What is the most likely diagnosis?

- A. Seborrheic keratosis
- ☒ B. Congenital compound melanocytic nevus
- C. Melanoma

Congenital CMN

CONGENITAL NEVUS

From Dr. Liisa Russell's Dermpath ppt

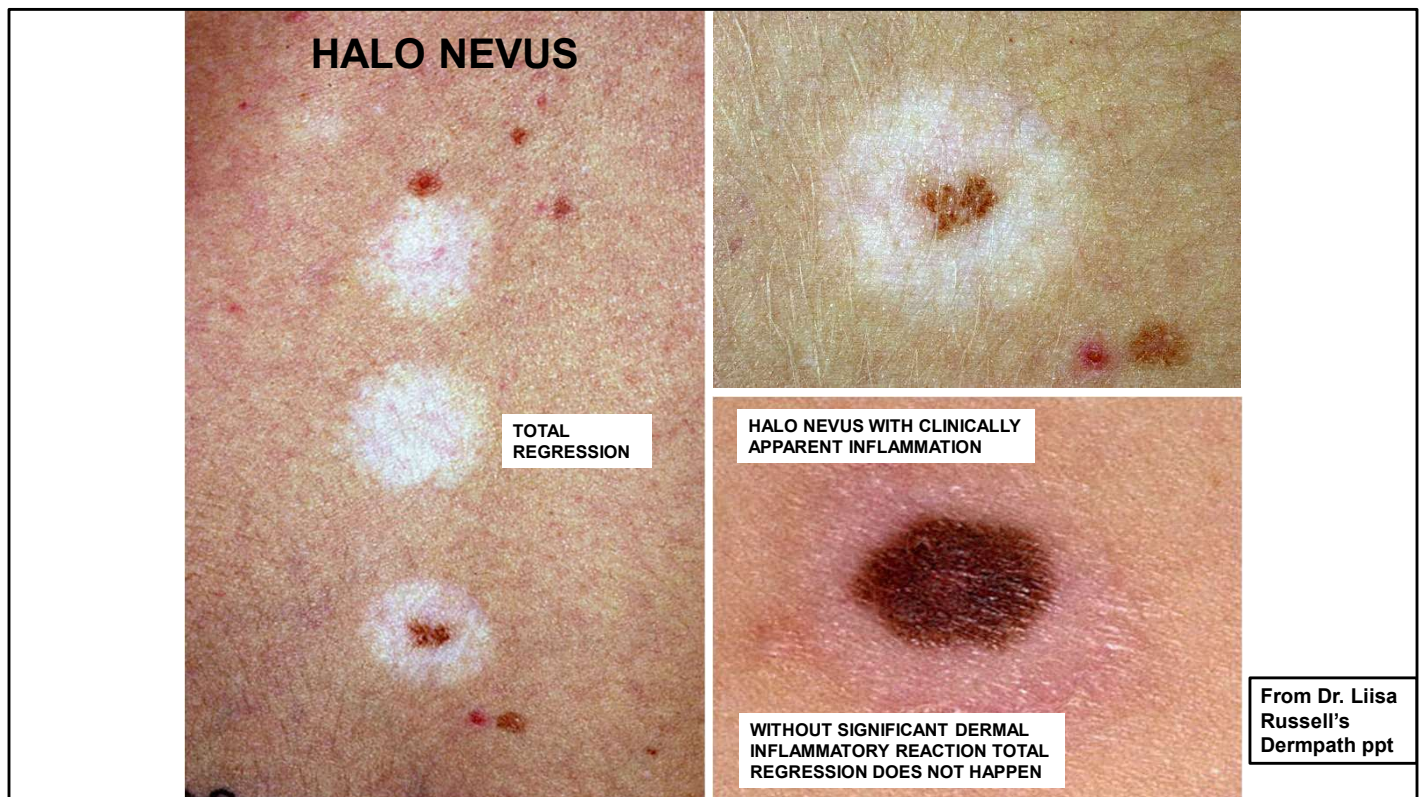


CLINICAL FEATURES:

- Present at birth; about 1% live births
- Often much larger than acquired nevi, measuring >1.5cm in diameter, occasionally >20cm
- Small congenital melanocytic nevi are usually raised and pigmented
- Those occurring on the scalp appear as convoluted, skin-colored "cerebriform" masses

GIANT CONGENITAL NEVUS:

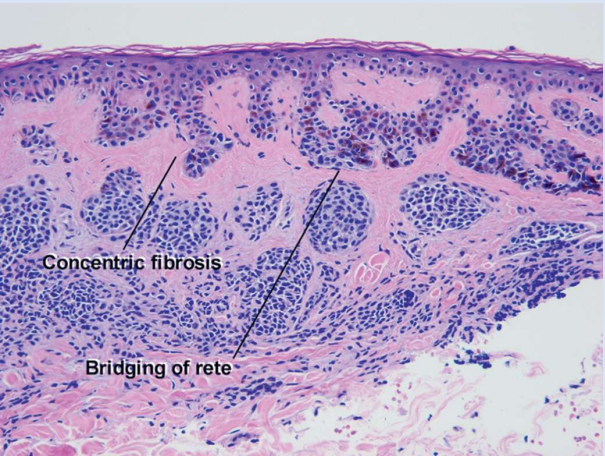
- May have a garment-like distribution (bathing suit, cap, stole, coat sleeve, or a stocking).
- Usually deeply pigmented and hairy with several satellite lesions.
- Leptomeningeal melanocytosis is often present if giant congenital nevus involves the neck or scalp.
- Blood vessels entering the spinal cord and brain are also surrounded by melanocytes.
- Large lesions (>20 cm) have about 6% lifetime malignant potential.
- Small lesions (<1.5 cm) rarely become malignant)



Halo nevus

white is used to be inflammation then regressed

not a nice homogenous color

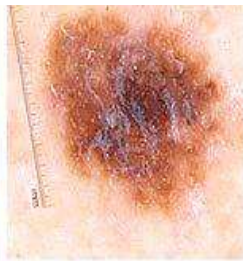
Dysplastic Nevus Melanocytic nevus that is clinically and histologically atypical appearing		
<u>Etiology and Pathology:</u> • Sun-exposure • BRAF mutated • Architectural and cytologic (cellular) atypia • "bridging" of melanocytic nests; fibrosis • Melanocytes show more hyperchromasia and increase in size		
<u>Clinical:</u> • >5 mm, can have irregular border, color variegation, "pebbly surface" • Classified as "benign" by the WHO as it's rare to transform to melanoma • But these are markers of individuals at increased risk for melanoma (3-20x) • Familial atypical multiple mole melanoma syndrome (auto dominant, >50 atypical nevi, 100% lifetime incidence of melanoma)	 Fig. 20.17B: Dysplastic nevus. This is often greater than 6 mm in diameter. Borders are typically irregular. From: The color atlas of the skin 10th Edition, 2015, the Institute of Dermatology, London, UK.	

- Melanocytic neoplasms
- Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 6, 101-131
-
- Dysplastic nevus
-
- Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.

DYSPLASTIC NEVUS SYNDROME



Dysplastic 1



Dysplastic 2



Dysplastic 3



Dysplastic Back 1



Dysplastic Back 2



Dysplastic Back 3

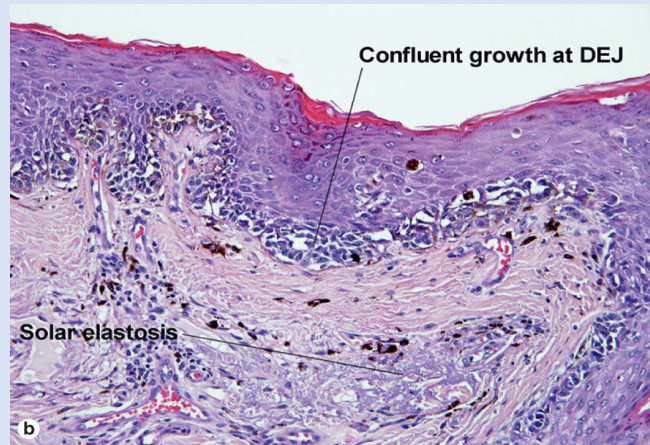
From Dr. Liisa Russell's
Dermopath ppt

Melanoma

Melanoma in situ refers to malignant melanocytes restricted to the epidermis; invasive melanoma refers to dermal infiltration

Etiology and Pathology:

- **UV light-induced DNA damage**
- Hereditary in 5-10% of cases
- **Radial and vertical growth phases**
- **Types of melanoma:**
 - **Superficial spreading melanoma (~70%)**
 - **Lentigo maligna melanoma (~5-10%; head & neck since associated with high degree of cumulative sun damage; older population than SS type)**
 - **Nodular melanoma (~15-30%; vertical growth phase melanomas)**
 - **Acrall melanomas (<5%)**



nodule >1 cm

Melanocytic neoplasms

Elston, Dirk M., Dermatopathology: Foundations in Diagnostic Pathology, Chapter 6, 101-131

Lentigo maligna (DEJ, dermal–epidermal junction)

Copyright © 2019 © 2019, Elsevier Limited. All rights reserved.

Melanoma

Tumor thickness (Breslow thickness) correlates with worse behavior



Clinical:

- "It is vital to recognize melanomas and intervene as rapidly as possible." (Robbins & Kumar Basic Pathology, 22)
- **Most lesions arise in skin but can also occur in the oral and anogenital mucosal surfaces, GI, meninges, and eye**
- Important clinical warning signs:
 - **Change in color or size of a pigmented lesion**
 - Rapid enlargement of a preexisting nevus
 - Itching or pain in a lesion
 - Development of a new pigmented lesion during adult life
 - Irregularity of the borders of a pigmented lesion
 - Variegation of color within a pigmented lesion

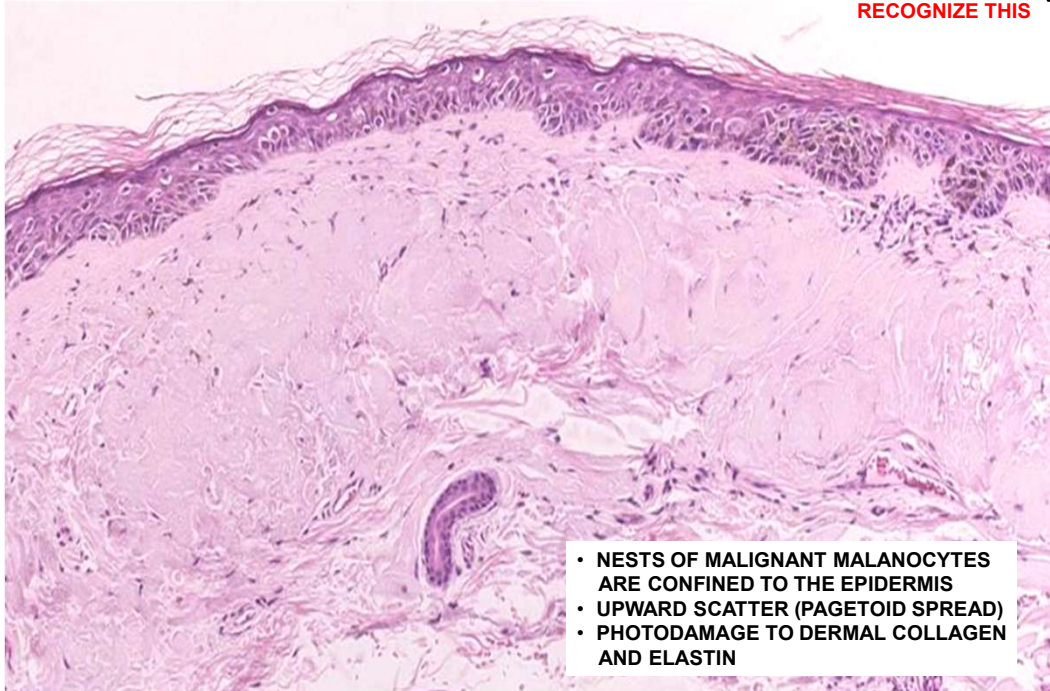
acral more common in people with more melanin bc has more protection

if feel something - do a lymph node exam

LENTIGO MALIGNA = MALIGNANT MELANOMA *IN-SITU*

From Dr. Liisa
Russell's
Dermopath ppt

RECOGNIZE THIS



- NESTS OF MALIGNANT MELANOCYTES ARE CONFINED TO THE EPIDERMIS
- UPWARD SCATTER (PAGETOID SPREAD)
- PHOTODAMAGE TO DERMAL COLLAGEN AND ELASTIN

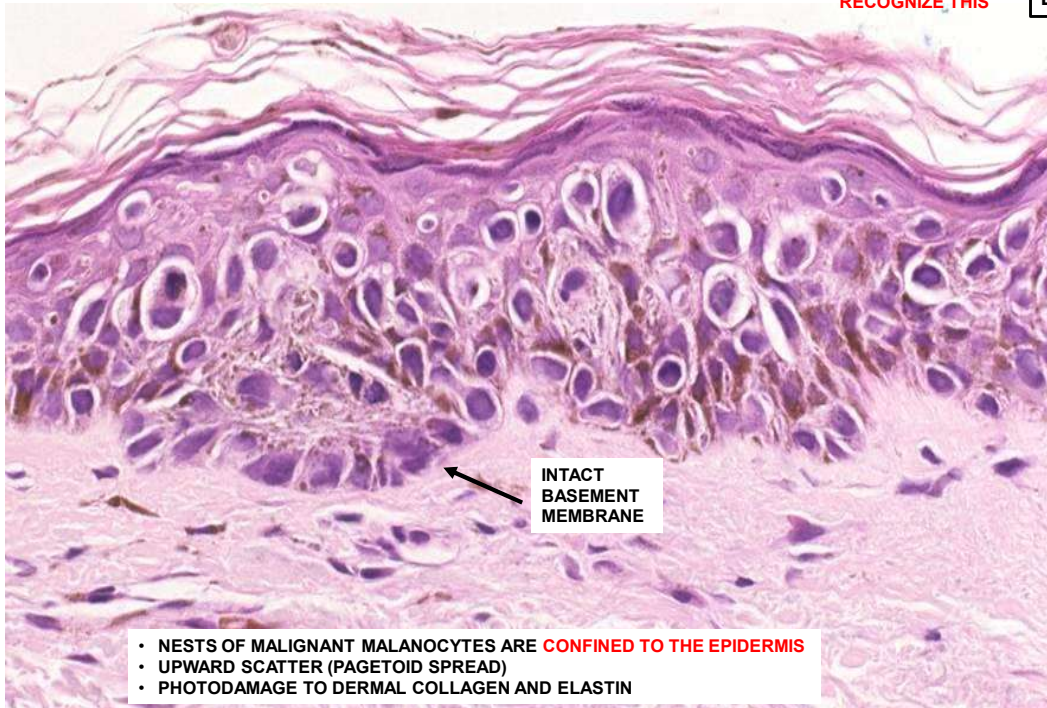
Melanoma in-situ. No dermal involvement.

irregular nuclear contours and enlarged

LENTIGO MALIGNA = MALIGNANT MELANOMA *IN-SITU*

RECOGNIZE THIS

From Dr. Liisa
Russell's
Dermopath ppt



69

85 yom, Malignant melanoma in-situ , nose. No dermal involvement, but atypical melanocytes throughout the epidermis.

LENTIGO MALIGNA MELANOMA

CLINICAL FEATURES:

- Large, macular, unevenly pigmented, poorly circumscribed lesion
- Occurs on sun-damaged skin

PATHOPHYSIOLOGY:

- Preceded by lentigo maligna (melanoma in-situ), which may be present for many years before invasive melanoma develops
- Long radial growth phase; lesion may reach up to 5-7cm in diameter before invasion of the dermis begins
- Nodular areas form as the tumor becomes invasive



From Dr. Liisa
Russell's
Dermopath ppt



Lentigo maligna melanoma, nose, 85 y/o man.

SUPERFICIAL SPREADING MELANOMA

CLINICAL FEATURES;

- Most common type; 70% of cases
- Most common locations lower extremities and back
- Peak incidence in the 4th and 5th decade
- Increasingly observed in young adults

PATHOPHYSIOLOGY:

- Has a relatively long phase of radial growth before penetrating deeper into the dermis and assuming the vertical growth phase.
- Flattened plaques with marked color variation and irregular margins.
- Partial regression causes pigment loss.
- May bleed.



From Dr. Liisa
Russell's
Dermopath ppt

NODULAR MELANOMA

CLINICAL FEATURES

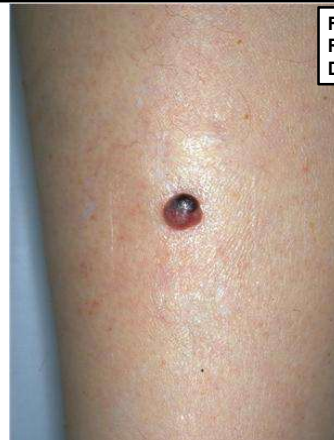
- Usually large, >6 mm; most are 10mm at diagnosis
- Asymmetric
- Poorly circumscribed
- Variegated color

PATHOPHYSIOLOGY

- Starts de novo, without radial growth phase
- Aggressive
- Metastases often present at the time of diagnosis

HISTOLOGIC FEATURES

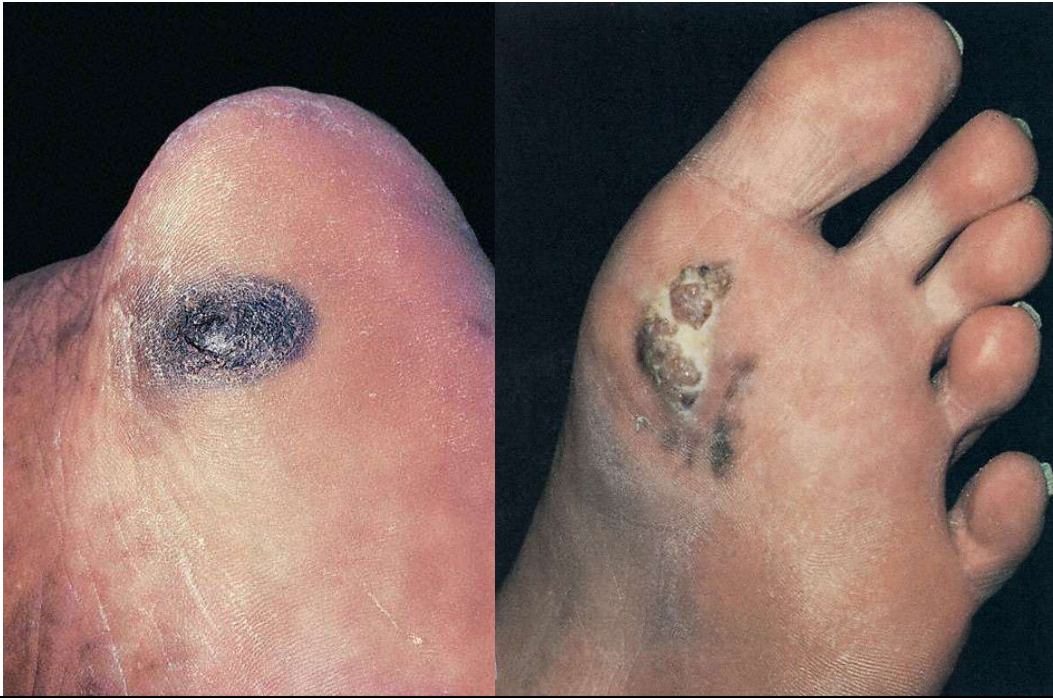
- No lateral spread
- Downward invasion into the dermis
- Upward infiltration into the upper epidermis may not be conspicuous
- Prominent cytologic atypia
- Dermal mitotic figures



From Dr. Liisa
Russell's
Dermopath ppt

GALLERY OF ACRAL LENTIGINOUS MELANOMAS

From Dr. Liisa Russell's
Dermopath ppt



Acral lentiginous melanoma

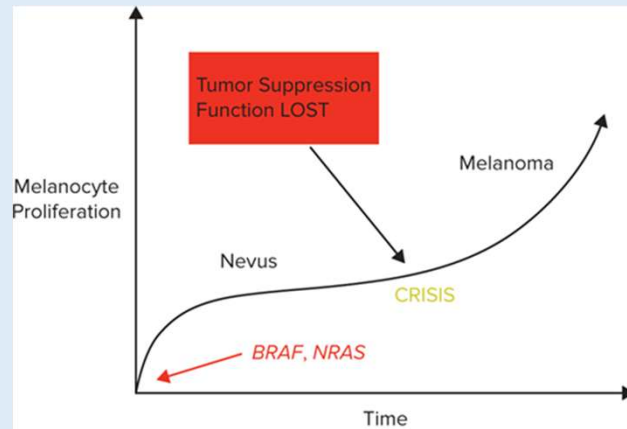
GALLERY OF ACRAL LENTIGINOUS MELANOMAS

From Dr. Liisa
Russell's
Dermopath ppt

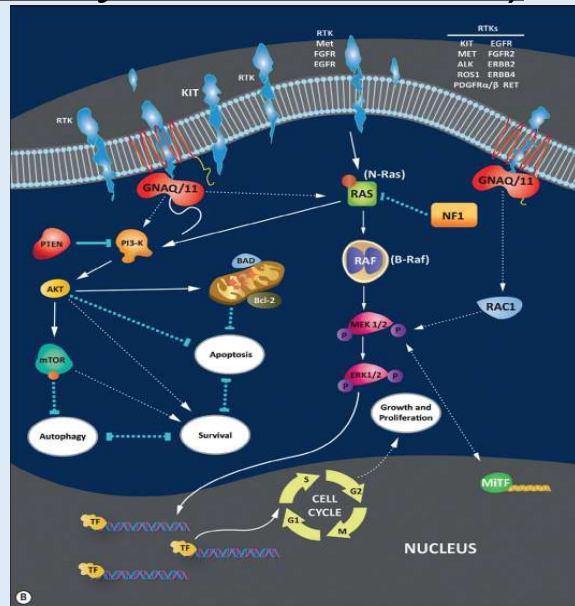


Acral (subungual) lentiginous melanoma

(FYI: Molecular pathways of melanoma)



Source: Garrett T. Desman, Raymond L. Barnhill: *Barnhill's Dermatopathology Challenge: Self-Assessment & Review*
Copyright © McGraw-Hill Education. All rights reserved.

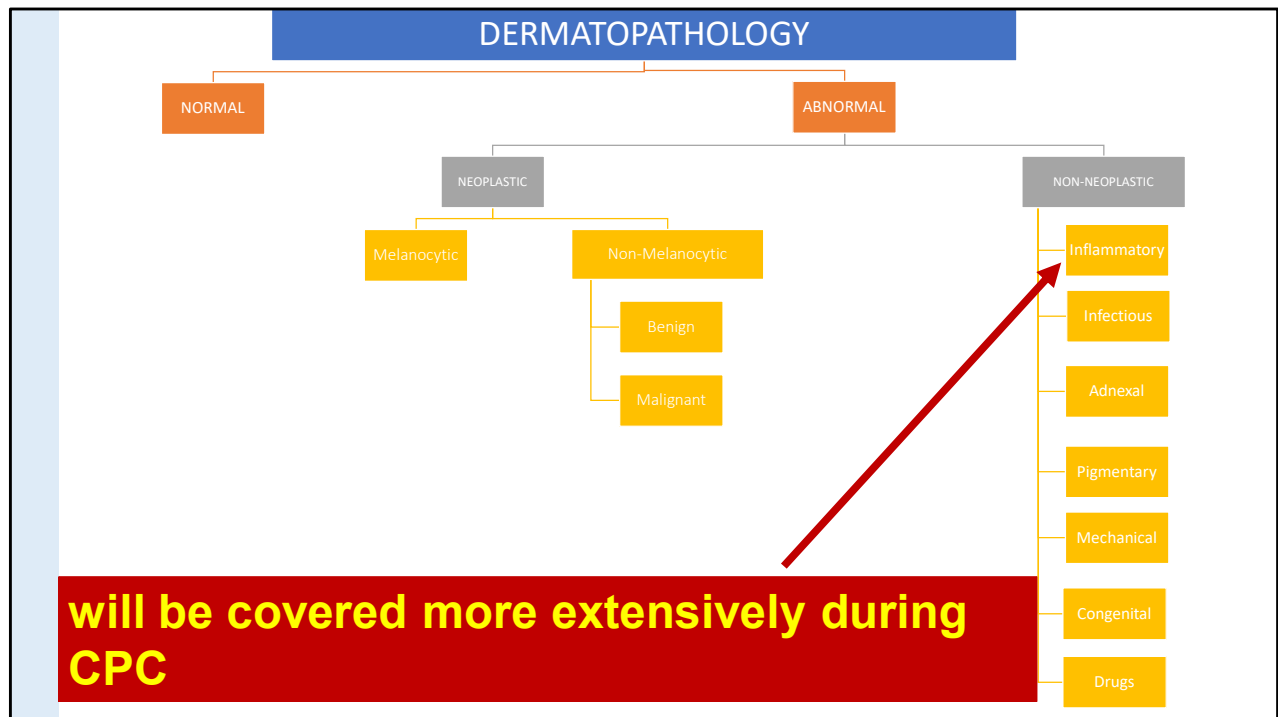


Weedon's Skin Pathology. Patterson, JW. 881-950

Molecular pathways of melanoma.

(B) The pathways that are more commonly altered in sporadic melanoma. *White arrows* are activation pathways. *Blue lines* are inhibition pathways. *Dotted white lines* correspond to indirect an/or less understood interactions.

Melanoma arising within a preexisting nevus. Oncogenic mutations (ie, point mutations in BRAF or NRAS) drive melanocytic proliferation. Tumor suppressor function is initially intact and the cells enter senescence; however, at a later point in time tumor suppressor function is lost and cells continue dividing and eventually leading to crisis. Genomically stable clones eventually emerge as malignant melanoma.

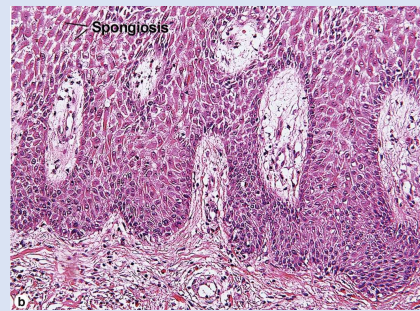
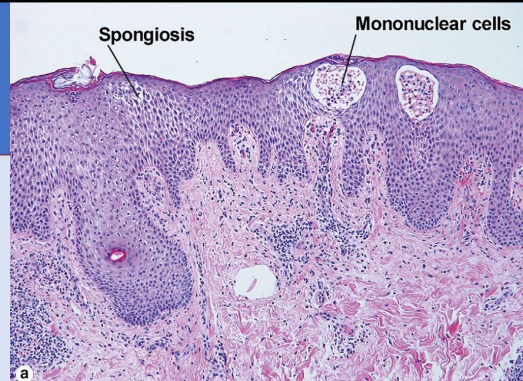


Spongiotic dermatitis

Aka eczematous dermatitis; intercellular edema

Etiology and Pathology:

- Endogenous dermatitis
 - Atopic
- Exogenous
 - Allergic contact dermatitis, irritant contact dermatitis



- Patterson, James W., MD, FACP, FAAD, Weedon's Skin Pathology

Psoriasis

Chronic inflammatory dermatosis. Parakeratosis, Munro microabscesses (neutrophil collections in the stratum corneum)

Clinicopathologic features:

- Psoriasis is an erythematous disease: erythematous because of dilation and increase in vessels in dermal papillae which have a thin overlying layer of keratinocytes
- Clinical thickening of lesions results from psoriasiform epidermal hyperplasia
- **Scale: parakeratotic cells resulting from increased transit time, and focal admixture of neutrophils**

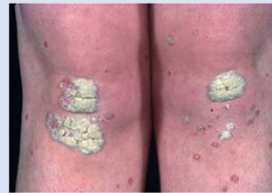
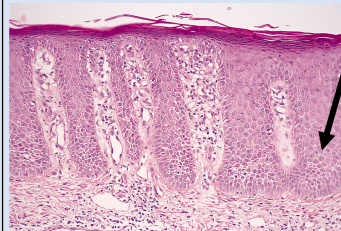
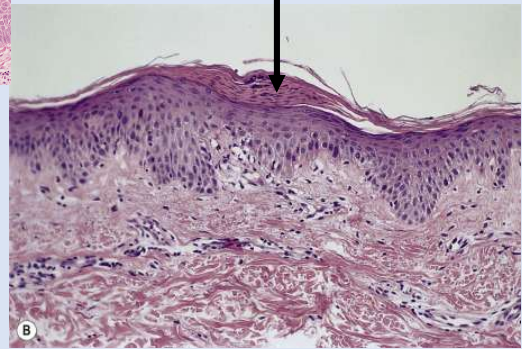


Fig. 6.67
Psoriasis: typical plaque disease showing bilateral and fairly symmetrical distribution. In this example, the silvery scale is well demonstrated. From:

Psoriasiform (regular) hyperplasia

Munro microabscesses and parakeratosis



The psoriasiform reaction pattern

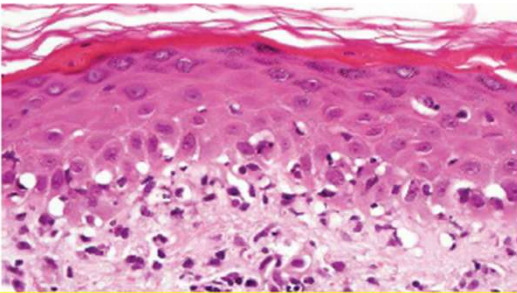
Patterson, James W., MD, FACP, FAAD, Weedon's Skin Pathology, 5, 99-120.e11

Psoriasis. (A) An early lesion with dilated vessels, perivascular lymphocytes, and exocytosis of lymphocytes and a few neutrophils. (B) A slightly later stage with neutrophils migrating to the summits of the parakeratotic mounds. (C) Dilated vesse...

Copyright © 2021 © 2021, Elsevier Limited. All rights reserved.

Interface Dermatitis

Interface Dermatitis



Basal keratinocyte hydropic change with vacuolization and variable lymphocytic inflammation

FYI

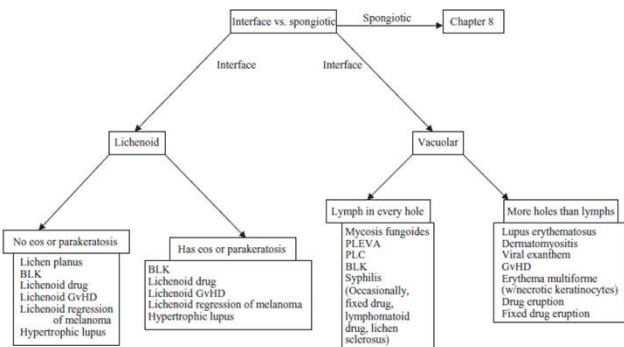


Fig. 7.1 Interface versus spongiotic dermatitis (EOS, eosinophils; PLC, pityriasis lichenoides chronica; PLEVA, pityriasis lichenoides et varioliformis acuta)

Photomicrograph from Dr. Kerri Rieger's Dermatology lecture series

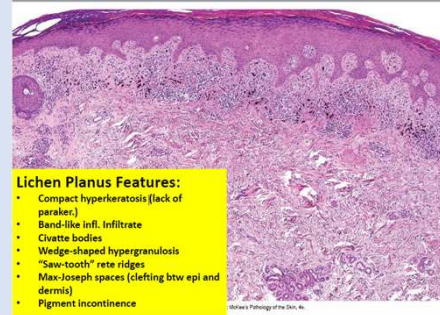
Interface dermatitis. Dirk M. Elston. Dermatopathology, Chapter 7

Lichen planus

Pruritic, purple, polygonal, planar papules and plaques

Clinicopathologic features:

- Unknown (CD8+ cytotoxic T cell-mediated response to unknown antigen being displayed by basal keratinocytes)
- Wickham striae are the fine white lines seen on surface of papules
 - Correspond to the hyperkeratosis and hypergranulosis



Photomicrograph from Dr. Kerri Rieger's Dermatology lecture series

Clinical photos from: North, Jeffrey P., McKee's Pathology of the Skin

THANK YOU!

study CPC and do the questions know why the other answers are wrong

what is cause of most cutaneous malignancies - UV

melanoma and nevi - BRAF

inflammatory disorders - can you describe them, mainstay is corticosteroids